

What Prices Can (and Can't) Tell Us About Cross-Border Shopping? Evidence from Argentina's Gasoline Market

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Abstract

This paper discusses the limits of relying on price data to detect and measure cross-border shopping (CBS), since a standard prediction is that the pass-through of a tax should vary with distance to the jurisdictional border. Using a novel station-level panel of prices and quantities for the universe of Argentine gasoline stations, I exploit a 2015 tax reform that expanded a southern tax-exempt zone and cut the applicable excise from 55% to 5%—a roughly 30% reduction in retail prices. Quantities sold rose 14% at exempted stations and fell 8% at non-exempted stations just north of the new border, with effects concentrated close to the border. Yet pass-through is spatially uniform within each region, indicating no price convergence despite substantial CBS. Five years later, survival probabilities diverge by roughly 10 percentage points across the border. I discuss the market structures that rationalize this gap between price and quantity responses.

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1 Introduction

Geographic borders create discontinuous changes in tax rates, which generate incentives for consumers to strategically travel across jurisdictions in order to benefit from lower prices. This behavior—commonly referred to as cross-border shopping (CBS)—has long been recognized in the economic literature, given its broad implications for public policy, and in particular for tax policy design. By enabling consumers to avoid higher-tax jurisdictions, CBS distorts tax collection by eroding the tax base, especially for local governments (Kanbur and Keen, 1993; Agrawal, 2012), and weakens the effectiveness of policies aimed at addressing externalities and market power (Christiansen, 1994; Christiansen, 2003). Beyond its fiscal implications, CBS has broader effects on local economies, influencing firms’ behavior, employment, and welfare (Campbell and Lapham, 2004; Baggs et al., 2016; Burstein et al., 2024).

Measuring the extent to which cross-border shopping (CBS) takes place is not trivial. Conceptually, CBS is an equilibrium outcome of a game that internalizes consumer, firm, and government responses (Agrawal, 2012; Christiansen, 2003; Kanbur and Keen, 1993). Much of the existing literature therefore exploits exogenous geographic variation or policy discontinuities to study how agents respond to changes in CBS conditions. Empirically, however, direct measures of cross-border consumption are rarely observed, leaving many studies to rely on an indirect test for CBS based on price convergence at jurisdictional borders. Following a tax increase in one region, reallocation of consumer demand toward a neighboring lower-tax jurisdiction should raise prices on the low-tax side and reduce them on the high-tax side. The spatial extent over which such price adjustments occur is informative about the geographic reach of CBS. Conversely, the absence of price convergence is often interpreted as evidence that existing price differentials are insufficient to offset travel costs, and thus that CBS is limited.

Using a novel dataset that combines station-level price and quantity data for the universe of gasoline stations in Argentina, this paper exploits an exogenous regional tax exemption that generated a sharp geographic discontinuity in gasoline excise taxes. The reform, enacted in December 2015, expanded a tax-exempt area within southern Argentina, reducing the applicable tax rate from 55% to 5% in newly covered regions, corresponding to a substantial decline in final prices of approximately 30%. The exogeneity and magnitude of this policy change allow me to study how consumers and firms adjust to changes in cross-border shopping conditions, and the longer run effects on the gasoline market structure.

To empirically investigate the immediate effects of this tax exemption, I construct a unique, geolocated monthly dataset at the gas station level containing average retail prices and total gasoline sales, based on administrative records from the Argentine Ministry of Energy. I divide the analysis into three parts. First, I examine the presence of cross-border shopping and firms’ pricing responses using a difference-in-differences approach that exploits variation across locations relative to the tax border. I compare gas stations

located south of the border, which benefited from the tax exemption, and stations located just north of the border, which were not directly affected by the reform, to a control group of stations located far from the border and thus plausibly unaffected by the policy change. I find that quantities sold at southern stations increased by 14% immediately and permanently following the tax reform, while prices fell by 24%. In contrast, northern stations near the border experienced an 8% permanent decline in demand, with little to no adjustment in prices. These results are consistent with price pass-through in tax-exempt areas and with consumer reallocation across the tax border.

Second, I assess whether price convergence at the border is a sufficient test for the presence of cross-border shopping by examining heterogeneity in price and quantity responses within the southern and northern regions. If CBS is driving the average effects, these responses should be concentrated among firms located closer to the border. I find that the positive and negative quantity effects in the southern and northern regions, respectively, diminish as stations are located farther from the border, consistent with consumer reallocation across jurisdictions. In contrast, price responses remain largely homogeneous within each region, regardless of distance from the border, providing no evidence of price convergence. These findings suggest that relying solely on price convergence may fail to detect CBS, even when substantial cross-border reallocation of demand is present.

Third, I examine the medium- to long-run effects of CBS on the number of firms in equilibrium, focusing on the exit rates of firms after the shock. I find that, consistent with the short-run results, the probability of operating in the market increases for firms located in the tax-exempted area relative to the non-exempted area. In particular, five years after the tax change, the probability of survival increases by 5% for the southern group and decreases by 5% for the non-exempted area relative to the control area. The results indicate persistent effects of the tax-induced demand reallocation on local market structure.

After establishing the short- and long-run effects of the policy, I turn to the question of which market structures can rationalize the joint pattern of significant CBS together with an absence of price convergence. I review several candidate models in which these two features coexist — and in which the standard price-convergence test would therefore fail — including competitive markets with constant marginal costs (Doyle and Samphantharak, 2008), settings where firms face consumers with heterogeneous demand elasticities due to the presence of mobile and immobile shoppers (Hindriks and Serse, 2019), and environments with price rigidities induced by market frictions, government regulation, or uniform pricing strategies adopted by multi-establishment firms (Friberg et al., 2022; Messner et al., 2024). I then contrast the predictions of each of these models with the full set of empirical results to rule out those that are inconsistent with the evidence and identify the mechanism that best explains the findings in this setting.

This paper contributes to the literature on cross-border shopping by studying how changes in CBS conditions affect regional price differentials and induce consumers to reallocate purchases across jurisdictions. Existing studies have typically measured CBS either

using aggregated data on consumption (Asplund et al., 2007; Banfi et al., 2005; Burstein et al., 2024; Campbell and Lapham, 2004; Chandra et al., 2014; Jansen and Jonker, 2018; Teixidó et al., 2024) or by relying on price convergence at jurisdictional borders as a test for CBS (Bajo-Buenestado and Borrella-Mas, 2019; Coyne, 2017; Doyle and Samphantharak, 2008; Hindriks and Serse, 2019; Louis and Taylor, 2016; Tsvetanov, 2024). This paper provides, to my knowledge, the first direct evidence of CBS using firm-level data that jointly observe prices and quantities on both sides of a border. This allows me to measure not only the extent of cross-border shopping, but also firms’ pricing responses to tax-induced demand reallocation.¹ Importantly, the results show that substantial CBS can arise even in the absence of differential pricing responses by firms across locations. More generally, the paper sheds light on the limitations of relying on heterogeneity in tax pass-through as evidence of the effects of taxation on firm and market outcomes.²

This paper also contributes to the literature on the long-run consequences of cross-border shopping for firm behavior and market structure. A large body of work studies how tax incentives around geographic borders shape firms’ location decisions (Bartik, 1985; Holmes, 1998; Peltzman, 2016; Rohlin et al., 2014). Related studies that analyze the long-run equilibrium effects of CBS typically abstract from variation in arbitrage conditions (Agrawal, 2015; Chiou and Muehlegger, 2008; Manuszak and Moul, 2009) or rely on temporary shocks, such as exchange-rate fluctuations, to generate changes in cross-border incentives (Baggs et al., 2016; Campbell and Lapham, 2004). By exploiting a large and sustained tax reform that sharply altered CBS conditions, this is the first paper to provide causal evidence on how changes in CBS patterns induced by a change in taxes reshape firm reallocation around jurisdictional borders.

The remainder of the paper is structured as follows. Section 2 provides institutional background on fuel taxation in Argentina, describes the 2015 TTLF tax exemption reform, and introduces the data. Section 3 presents the empirical methodology and short-run results. It begins with the difference-in-differences estimates of average price and quantity responses on each side of the tax border, before turning to the heterogeneity analysis that examines how these effects vary with distance to the border and discusses the implications for the price-convergence test. Section 4 analyzes the long-run effects of the reform on the probability of survival for firms around the border. Section 5 evaluates different mechanisms and market structures that can reconcile the results from previous sections. Section 6 presents a series of robustness checks, including heterogeneous effects by brand, the role of local competition, and a dynamic event-study analysis that validates the parallel trends assumption and rules out anticipatory behavior. Finally, Section 7 concludes.

¹An exception is Harding et al. (2012) in the context of cigarettes, which uses Nielsen transaction data and shows both lower tax pass-through near borders and an increased probability of out-of-state purchases following tax increases.

²For instance, several papers exploit pass-through variation to infer market power, tax evasion, regulatory constraints, or distributional incidence (Benzarti et al., 2024; Genakos and Pagliero, 2022; Kopczuk et al., 2016; Stolper, 2024).

2 Institutional Background and Data

This section provides institutional background on fuel taxation in Argentina and the structure of the gasoline retail market. I describe the Tax on the Transfer of Liquid Fuels (TTLF) and the 2015 reform that expanded tax-exempt areas. I then introduce the datasets used in the analysis and present descriptive statistics.

2.1 Institutional Background

Argentina is a federal country with three levels of government: the national government; the subnational level, comprising 23 provinces and the Autonomous City of Buenos Aires (CABA); and more than 2,300 local governments. Historically, fuels have been primarily taxed through a national excise tax known as the Tax on the Transfer of Liquid Fuels (TTLF). Prior to 2015, gasoline was taxed uniformly across the country at an ad valorem rate of 62%, a rate that had been in place since 2003. In January 2015, the national TTLF rate was reduced to 55.8%. In addition to the TTLF, gasoline was subject to other taxes during the period analyzed. Specifically, it faced two additional national taxes: (i) the value-added tax (VAT) at a rate of 21%; and (ii) an additional excise tax of 5% earmarked for financing the Hydric Infrastructure Fund³.

2.2 The 2015 TTLF Tax Exemption Reform

The TTLF includes an exemption zone in place since 1998, historically referred to as the *42nd parallel zone*. This zone covered the Patagonian regions in southern Argentina, specifically the provinces of Tierra del Fuego, Santa Cruz, and Chubut, as well as selected municipalities in Río Negro and Neuquén (see Figure A.1, original border in blue). The rationale for this exemption was to promote economic activity in remote and sparsely populated regions characterized by higher transportation and logistics costs.

On November 25, 2015, the exemption zone was extended to include three entire provinces—Río Negro, Neuquén, and La Pampa—as well as two municipalities in additional provinces: Patagones, located in the south of Buenos Aires Province, and Malargüe, in Mendoza Province (see Figure A.1, new border in red). As a result, gasoline stations located south of the newly defined border benefited from a full exemption from the TTLF. This reform reduced the applicable TTLF rate from 55.8% to 0%, corresponding to a decline in the final consumer price of -30.7% ⁴.

³To the best of my knowledge, there were no other excise taxes on gasoline at the state/province level. At the local level, some counties levy a tax on gasoline consumption but it is really low compared to the national tax (Porto et al., 2019). A discussion of the potential concerns around other taxes will be addressed in Section 3.1.

⁴The change in the tax-inclusive price includes the VAT and the Hydric Infrastructure Fund tax mentioned above and is calculated as follows: $\left(\frac{1+0.21+0.558+0.05}{1+0.21+0+0.05} - 1 \right) \times 100$

A notable feature of this tax reform is that it was not driven by contemporaneous market conditions. After more than twelve years in power, the incumbent government lost the presidential election on November 22, 2015. In the weeks preceding the transfer of power on December 10, 2015, the outgoing administration enacted a series of laws and regulations aimed at constraining future revenue streams available to the incoming government.⁵ The expansion of the TTLF exemption zone was part of this broader set of fiscal measures. Importantly, the reform was implemented immediately: three days after the election, the bill—previously approved by the Senate—was enacted by the federal administration and took effect on December 1, 2015. Table A.5 summarizes the legislative timeline of the tax change.

2.3 Market Structure and Pricing

The oil industry value chain can be broadly divided into two main segments: upstream and downstream. The upstream segment comprises exploration and production activities, while the downstream segment includes refining—the conversion of crude oil into fuels for energy consumption—distribution, and retail sales at the gas station level. In Argentina, the refining sector is highly concentrated, with five firms accounting for more than 95% of total refining capacity. Among these, the state-owned company YPF controls over 50% of national refinery capacity.⁶ After refining, fuel is transported to dispatch terminals, where it is stored before being delivered to gas stations by tanker trucks. Each major refinery operates its own delivery fleet, which constitutes the primary mode of fuel transportation in the country (Scheimberg, 2018). Figure A.13 illustrates the structure of the oil value chain.

At the retail level, Argentina has approximately 5,600 gas stations, over 85% of which operate under a branded network. Major refining companies maintain extensive retail networks, either through direct ownership or through majority ownership stakes. Although regulation limits ownership to no more than 40% of stations operating under a given brand,⁷ contractual arrangements are asymmetric, allowing refineries to exert substantial influence over pricing policies and operational decisions (Scheimberg, 2018). The remaining stations operate under a “white flag” arrangement, analogous to unbranded gas stations in the United States.

2.4 Data and Descriptive Statistics

The empirical analysis relies on a main dataset constructed using open administrative records from the Argentine Ministry of Energy. I am able to build a geolocated monthly

⁵For related evidence, see contemporaneous reforms to the pension system and changes to the redistribution of federal funds to provinces, like the one published in this *link*.

⁶The remaining capacity is held by Shell (16%), Axion (14%), Oil Combustibles (8%), and Petrobras (5%).

⁷Decree 1060/2000.

panel containing retail fuel prices and total fuel sales for the universe of gasoline stations in Argentina for the period 2008 -2022 . The dataset reports the total quantity sold and the average monthly retail price for each fuel type,⁸ the brand affiliation of each gasoline station, and its geographic coordinates.

I focus on regular and premium gasoline, as these are the most consumed fuels by final consumers, representing more than 60% of total quantities sold. As in the United States, the distinction between regular and premium gasoline primarily reflects differences in octane content, with premium gasoline having higher octane levels and a higher retail price. Other fuel types include diesel, kerosene, and liquefied petroleum gas (LPG). Diesel is widely used by trucks and agricultural machinery and is not subject to the TTLF when consumed as an intermediate input. Although the tax is collected at the pump, firms can later deduct it as an advance payment for other taxes. Kerosene and LPG represent a small share of total fuel retail sales.

Table A.1 presents descriptive statistics for the universe of 5,601 gas stations over the period 2008 -2022 . On average, each gas station appears for about 141 months (≈ 11 years). Panel A shows that each gas station sells around 117.5 thousand liters per month of regular gas and 48.6 thousand liters of premium gas (equivalent to 31.0 and 12.8 thousand gallons, respectively). For reference, the California Annual Retail Fuel Outlet Report (CEC-A15) estimates that, on average, a gas station sold around 88 thousand gallons per month in 2023. Because of the inflationary context, prices are presented in real pesos of December 2015. On average, a liter is about 14.8 pesos per liter for regular gas, equivalent to 4.9 dollars per gallon as of December 2015. Panel B in Table A.1 describes the distribution of stations for the period. As noted in Section 2, the government-owned brand stands out by the number of gas stations that operate under its name, and market share, both in regular and premium gas. The second brand is Shell, and the third in number of stations are *White Flags*, independent gas stations that operate under no brand name, similar to unbranded stations in the US.

Station-level geographical coordinates are used in two ways. First, they allow me to measure proximity to the new tax border, by computing, for each station, the distance to the nearest competitor located on the opposite side of the border. Throughout the paper, each such distance is computed in two ways: (i) linear Euclidean distance using the *geodist* package—developed in Picard (2010); and (ii) route-based driving distance (kilometers along the road network and travel time in minutes) using the *osrmtime* package—developed in Huber and Rust (2016)—with the free Open Street Map OSRM software⁹. Figure A.15 summarizes how the distance to the nearest station on the other side of the border is distributed when computed as straight-line (Euclidean) separation versus as route-based

⁸The price and quantity data are available at <http://datos.energia.gob.ar>.

⁹The package has advantages over those that use online route distance services. First, georeferenced data cannot always be used online due to sensitivity issues. Additionally, offline routines using free software are not at risk of becoming obsolete due to online software updates. Finally, since the results do not depend on time-specific real-time data, they can be replicated at any time.

driving distance along the road network. Second, coordinates allow me to measure distance between stations in order to describe the local competitive environment. Panel C of Table A.1 reports the average number of competitors within a given radius around each station, using the same Euclidean and route-based definitions.

The dataset is novel in several respects. First, it contains quantity information at the firm level. Papers that analyze CBS with quantity data generally rely on aggregated regional data (Asplund et al., 2007; Bajo-Buenestado and Borrella-Mas, 2019; Banfi et al., 2005; Teixidó et al., 2024) or consumer expenditure surveys (Burstein et al., 2024; Chandra et al., 2014). Second, the precise geolocation of each station allows for the calculation of the true distances to the nearest competitor, complementing previous papers using other proximity measures such as zip code centroids or population centers (Agrawal, 2015; Harding et al., 2012). Third, the data covers the universe of gasoline stations, complementing previous studies that rely on sample of firms—often collected manually (Manuszak and Moul, 2009) or through credit card data sources such as OPIS or GasBuddy (Coyne, 2017; Doyle and Samphantharak, 2008; Louis and Taylor, 2016; Tsvetanov, 2024). Finally, the dataset spans a long period of time, making it possible to observe firm entry and exit without additional assumptions.

3 Short Run Effects

3.1 Methodology

The strategy to study the effects of the tax change in the area around the border consists of employing a difference-in-differences approach. However, instead of using two groups (treatment and control), I will define three: a control group composed of gasoline stations located far from the border—assumed to be unaffected by the tax exemption—and two treatment groups. The first treatment group includes stations located south of the border, which benefited from the tax exemption, while the second group includes stations located just north of the border, which were not directly affected by the tax change. The main model to estimate is equation 1:

$$Y_{i,t} = \alpha + \beta_S \times South_i \times Post_t + \beta_N \times North_i \times Post_t + \eta_t + \psi_i + \epsilon_{i,t} \quad (1)$$

where $Y_{i,t}$ is the log of price or quantity in gas station i during month t . $South_i$ ($North_i$) is a binary variable equal to 1 if the station i is located within 175 km (≈ 108.7 miles) south (north) of the border. The control group consists of firms that are located north, far away from the border 550 – 750 km (342 – 466 miles). A fixed time effect η_t is included to control for seasonal patterns common to every gas station. Fixed effects ψ_i for gas stations are included to control for any time-invariant characteristics. Errors are clustered at the county level. Because it is interacted with a binary variable, the coefficient's transformation $e^{\beta_S} - 1$ measures the percentage change in the southern area compared to the control group

after the tax reform. The analogous interpretation applies to β_N for the northern group.

The key identifying assumption is that, absent the treatment, average price and quantity in both southern and northern stations would have followed the same trend compared to the control group over time. Figure A.2 presents descriptive evidence supporting the former by plotting the evolution of the trend for the three groups for premium gasoline. Panel (a) shows that, prior to the reform, average prices were similar across all groups. After the reform, exempted firms (blue line) were able to lower prices significantly. In contrast, stations on the northern side of the border (red line) displayed a price trend similar to that of the control group (green line). For quantities, panel (b) shows the similar trends for the three groups before the reform, and the divergence after December 2015, where quantities increased for southern stations and declined for northern ones compared to the control group. Given the pre-trends, it would be difficult to argue a divergence in outcomes after the reform that is not related to the tax change that happened in December 2015¹⁰.

Another important assumption is the absence of significant anticipation effects before the policy is implemented. In this context, anticipation effects would take the form of consumers delaying their consumption until the tax cut gets into implementation (Coglianese et al., 2017). Firms could also react by lowering prices in the south before the reform and increasing them after the reform in response to excess demand. The nature of the reform, attached to the political cycle, makes the anticipatory behavior implausible. The reform was passed two days after losing the elections and fully implemented 10 days after the former. Even if some consumers would have delayed consumption until the full implementation of the tax cut, there exists a physical constraint for intertemporal substitution. The maximum amount a vehicle could have stopped consuming and shifted to the following period after the reform is only one tank. It does not seem very likely these anticipatory behaviors could last more than a few days or have the strength to bias the estimates. Anyway, the availability of monthly data will address these effects which should be concentrated in the months around the reform. Section 6.3 will deal with dynamic effects and show no evidence of anticipatory behavior.

The coefficient β_s identifies not only the effect of CBS, but also the own price elasticity of demand. Lowering the price of gas might also encourage a genuine increase in demand for gas, not related to CBS. The sum of these two effects will be captured by β_s . If the elasticity is relatively high, then that will bias the coefficient upwards. This is less of a concern in this setting because first, the literature finds a low elasticity of gas in general¹¹. Second, the availability of quantity data on both sides of the border allows to test if quantities are increasing in the south and decreasing in the north at the same time, consistent with CBS. On the north side, this concern will be less of an issue, unless there is a significant price

¹⁰Regular gas evolution is depicted in Figure A.18. Although quantity levels differ, with the control group starting from a higher quantity sold, the parallel treatment assumption holds and follows the same pattern after the tax change.

¹¹At least at the aggregate level, see Levin et al. (2017) for a review on the subject and a discussion of how the elasticity can vary at the firm level compared to the aggregate level.

response. The heterogeneity analysis in Section 3.3 will help to address this identification concern, as there should be heterogeneous effects depending on distance to the border.

A further concern for the southern region is the *original* tax border established in 1998 (Figure A.1). That border created incentives of the same kind as the new one, but in the opposite direction: before the reform, the southernmost region remained exempt, so firms there enjoyed a favorable tax differential and arbitrage ran from north to south within the affected areas. The reform eliminates that differential along the original line, which in principle could shift consumption from south to north of the old border—where the treated group *South* is defined. Two identification issues follow. First, stations on the southern side of the *new* border could be contaminated by reallocation tied to the disappearance of the old border, biasing estimated effects upward; therefore results for stations located too close to the original border should be interpreted with caution. Second, and more serious, if some stations classified as “south” in the distance window actually lie south of the original border, the same coefficient would mix the new-border treatment with negative effect on firms on the southern side of the old line. To avoid that, I restrict the sample to gas stations located north of the original border.

In principle, the change in incentives around the original border would allow an analogous analysis with a separate indicator for stations on the southern side of that border. This paper focuses on the new border because the goal is a counterfactual of firms that competed under similar conditions and were then shocked by a location-based rule that was not tailored to the ex-post tax differential they would face. The old border, by contrast, is associated with a long distortion that plausibly induced reallocation toward the south of the original line—as suggested by Figure A.1—and station density is lower there, which would further complicate power problem in such estimation.

In this paper I am not addressing how local governments react to the federal tax change, assuming implicitly that they are not responding to CBS. A central debate in the CBS literature is how local governments set optimal taxes in response to CBS (Agrawal, 2012, 2015; Christiansen, 1994, 2003; Kanbur and Keen, 1993; Ohsawa, 1999). If local governments are reacting to CBS, the interpretation for the price change would be then a sum of firm and local government responses. This is not a concern for the quantity estimation as I am using the final retail price at the pump, which is the one that consumers observe and pay. A future line of research would be to analyze how local governments respond to the tax change by adjusting local taxes. At the moment, I am not aware of any homogenous dataset that collects local taxes set by local governments over different periods in time. Even if in theory local taxes might be adjusting, the total effect on prices would be small given that local taxes are really small compared to federal ones¹².

Another caveat in this setting is the entry and exit of firms. If because of the tax cut, a significant number of firms enter the market in the south, the coefficient would be biased towards zero, as the quantity demanded would then be divided by more firms. On the other

¹²See Porto et al., 2019 for a discussion on the local taxes in the province of Buenos Aires.

side of the border, the inverse would happen. Given that, in the short run, reallocation seems unlikely because of the (literally) sunk costs associated with each gas station, I will focus this analysis on the window around the tax reform 2014 -2017 , using only a balanced panel of firms that do not enter or exit the market in that period. The long-run duration would be a concern if the analysis included firms that entered or exited the market in the period, but the analysis is restricted to a balanced panel of firms that do not enter or exit the market in that period. The long run discussion will be deferred for Section 4.

The final sample is described in Table A.2. There are 87 stations that receive the tax exemption (and are close to the border), 154 stations close to the border that do not receive the exemption, and 564 stations located in regions far from the border, which I assume are not affected by the tax change. Panel (A) in Table A.2 shows the difference in quantities sold by the three groups, where the control group sells around 21 and 38 thousand liters more compared to the south and north groups, respectively. On the other hand, the three groups set virtually the same price, both for regular and premium. Panel (B) in Table A.2 presents the distribution of stations. There is some overrepresentation of YPF brand in the area around the border. The explanation comes from underrepresentation of Shell stations, whereas the white flags have a constant share in the three regions. Panel (C) in Table A.2 shows that the number of nearby stations is fairly similar in the three groups, especially for the smaller radius, even though there exists important variance for each of the different distance variables within each group.

3.2 Average pass-through and quantity change in the area around the border

Table A.3 shows the results of estimating Equation 1. Prices fall after the reform in the exempted region by 21.8% and 21.8% for regular and premium gas, respectively. Given the tax structure, prices drop 8.9 percentage points less than they would with full pass-through, represented by a drop of 31% . The implied pass-through of 70.9% is in line with the lower bound of the literature, which estimates a range from 70% to close to unity (see Alm et al., 2009; Chouinard and Perloff, 2004; Genakos and Pagliero, 2022; Marion and Muehlegger, 2011).

The effect on prices in the non-exempted region is theoretically ambiguous. If demand is shifting towards the south because of CBS, prices might decrease in the non-exempted area — a phenomenon known in the literature as *cross-border pass-through* (Bajo-Buenestado & Borrella-Mas, 2019). But prices could also rise in the non-exempted region if there is a change in demand elasticity for the northern stations (Hindriks and Serse, 2019). To illustrate this case, assume two types of consumers, mobile and non-mobile shoppers, with the former having lower travel costs and a higher elasticity of demand. If, after the tax, most of the mobile shoppers go to the southern region to purchase gas and the northern firms face mostly inelastic consumers, then the price will tend to rise in the north. The equilibrium price would then be the result of two forces: a lower demand that pushes the

price downwards but a more inelastic one that pushes it up. If the two effects offset, the result is zero change in prices — paradoxically the same outcome as in a situation where CBS does not take place at all. Empirically, β_N has positive point estimates, even though the magnitude is 25 times smaller than β_S and not always statistically significant¹³.

The price results are uninformative about whether CBS is taking place. Given the drop in prices in the exempted area and virtually no change in the north, the natural next step is to examine quantities sold on each side of the border, since CBS is fundamentally about consumers shifting their purchase decisions. The estimates are presented in Table A.4, where the specification follows Equation 1 but using the log of quantities as the dependent variable. Quantities increase in the exempted region by 15.3% for regular and 41.0% for premium gas. Combined with the price drops estimated above, a naive calculation of own-price elasticity would imply values of 0.70 for regular and 1.88 for premium gasoline — orders of magnitude higher than typical estimates of gasoline demand elasticity (Brons et al., 2008; Coglianese et al., 2017; Hughes et al., 2008; Levin et al., 2017; Li et al., 2014).

Under CBS, the southern coefficient will capture both the demand response of local consumers and the reallocation of buyers from the north. If consumers are crossing the border, quantities in the non-exempted region should fall as those same buyers shift their purchases to the south. The coefficient β_N on quantities provides a direct test of this hypothesis. Table A.4 confirms the prediction: quantities in the non-exempted area decrease by 7.3% for regular and 11.3% for premium gasoline. The effect is stronger for premium gasoline on both sides of the border.

These results complement a literature on CBS responses to exogenous price differentials — typically driven by taxes or exchange rate movements — which has relied on either regional aggregates (Asplund et al., 2007; Banfi et al., 2005; Leal et al., 2010; Teixidó et al., 2024) or consumption surveys (Burstein et al., 2024; Chandra et al., 2014; Jansen and Jonker, 2018). The geolocated, firm-level quantities allow me to measure CBS directly in a homogeneous region where no prior tax differential existed. Whether prices and quantities respond heterogeneously with distance to the border is taken up in the next subsection.

3.3 Heterogeneity, price convergence and distance to the border

If consumers are crossing between regions, then firms closer to the border might internalize this behavior and will be the ones with more incentives to adjust their prices. Price convergence at the border has been used as an indirect test for the existence of CBS: prices on the two sides of the border should converge as nearby firms compete for the same mobile consumers, while firms farther away — facing more captive demand — should respond less (Bajo-Buenestado and Borrella-Mas, 2019; Coyne, 2017; Doyle and Samphantharak, 2008; Tsvetanov, 2024). The empirical prediction is therefore a price gradient with respect to distance to the border.

¹³See for example Figure A.12 for different definitions of the treatment groups.

I will test that prediction by examining heterogeneity in the responses with respect to distance to the border. Disaggregating by distance serves three purposes. First, it tests whether the quantity effects identified above are driven by CBS, since under CBS sales should be concentrated among stations closer to the border. Second, it tests whether the seemingly low price response in the North is a precise null or reflects convergence in a narrow band at the border that regional averages cannot detect. Third, it sheds light on how far the effects of CBS reach from the border.

I divide gas stations into bins based on their distance to the tax border, where for each station the relevant distance is to its nearest competitor located on the opposite side. To estimate heterogeneous effects by distance, I run Equation 2:

$$Y_{i,t} = \alpha + \sum_{k=1}^B \beta_k \times Post_t \times I_i\{k\} + \eta_t + \psi_i + \epsilon_{i,t} \quad (2)$$

where $I_i\{k\}$ is an indicator equal to 1 if station i falls into bin k . The total number of bins is B , with an equal number on each side of the border, and the control group consists of firms located 550 – 750 km (342 – 466 miles) from the border, as before. Equation 1 is the special case in which $B = 2$. The identifying assumptions are identical to those in Section 3.1.

The number and length of bins involves a trade-off: more bins imply narrower regions, which can identify more point estimates at the expense of a smaller number of stations per bin, generating noisier estimates. The main specification uses 5 bins of width 35 km ($\approx$ 22 miles) on each side. These bins are wider than what the literature typically treats as a geographical market (Davis et al., 2025; Stolper, 2024; Taylor and Muehlegger, 2025), reflecting the lower station density of this rural region (Figure A.15). Robustness to alternative bin widths is reported in Figures A.16 and A.17; the main conclusions are unchanged.

The results of estimating Equation 2 are presented in Figure A.3. The blue estimates show the percentage change in prices for each bin following the policy. Consistent with the average results, firms in the exempted area exhibit lower prices after the tax change while prices remain virtually unchanged in the non-exempted area. Within each region, however, the price response is homogeneous regardless of firms' locations¹⁴: no visible price gradient appears in either the southern or northern bins. The convergence prediction fails to hold — even at narrower intervals, the price response within each region is homogeneous regardless of firms' locations.

In contrast, the quantity estimates display a clear distance gradient. The red estimates in Figure A.3 report the quantity responses for both fuel types. In the exempted region, the quantity increase is concentrated among stations closest to the border and falls off with

¹⁴A formal test of homogeneity by distance is presented and discussed in Table A.10 for different bin definitions.

distance¹⁵. The same pattern appears on the non-exempted side: quantities decline most sharply close to the border and attenuate with distance, though significant effects persist as far as 175km ($\approx$ 115 miles) from the line. CBS therefore extends well beyond stations immediately adjacent to the border.

Taken together, these results document a setting with strong cross-border reallocation of quantities alongside no price convergence at the border. To the best of my knowledge, this is the first paper to document this combination of patterns identified from a tax change within an institutionally homogeneous region. A key feature that makes the evidence possible is the joint observation of prices and geolocated quantities at the firm level. These results add to a literature that has identified CBS through quantity data by bringing firm-level pricing into the same setting, allowing the price and quantity responses to CBS incentives to be examined jointly (Asplund et al., 2007; Banfi et al., 2005; Burstein et al., 2024; Campbell and Lapham, 2004; Chandra et al., 2014; Jansen and Jonker, 2018; Teixidó et al., 2024). Two closely related studies combine information on prices and quantities. Harding et al. (2012), in the context of cigarettes, uses Nielsen-level transaction data to show that the pass-through of taxes is lower close to the border and that the probability of buying out-of-state rises with higher taxes. In contrast to the monotonic gradient documented above, Friberg et al. (2022) documents a non-monotonic effect of distance on quantities in response to exchange rate changes in grocery stores along the Norwegian-Swedish border.

4 The Long Run: Probability of Survival After the Tax Shock

The short-run results reveal an asymmetric adjustment: quantities fall sharply in the northern, non-exempted region while they rise in the south, yet northern firms display virtually no price response to the adverse demand shock they face. If northern stations cannot recover lost sales through reducing prices their revenue will be structurally lower than it was before the reform. Faced with persistently less quantities, exit might arise as the margin of adjustment. Symmetrically, southern stations experiencing a significant volume increase may attract new entrants seeking to capture the rents generated by the tax exemption. A natural question, therefore, is whether the price differential is strong and persistent enough to translate into a long-run reallocation of firms across the border. This section evaluates the probability of survival of firms in each region from two years after the policy onwards.

To test if the tax change changed the probability of survival of a firm, I estimate a linear probability model based on Equation 3:

¹⁵Results for the southernmost bins should be interpreted with caution. Figure A.3 shows a quantity increase in those bins that is likely driven by the presence of the old border, as discussed in Section 3.1.

$$D_{i,t} = \alpha + \sum_{t=0}^T \beta_S^t \times South_i \times I_t\{t\} + \sum_{t=0}^T \beta_N^t \times North_i \times I_t\{t\} + \eta_t + \epsilon_{i,t} \quad (3)$$

where $D_{i,t}$ is a binary variable equal to 1 if the firm i is operating in the market in period t . The periods used are every month in between 2008 -2022 . The sample consists of the balanced set of firms active in the market in the short-run analysis. For the periods after the reform, the coefficient β_S^t is interpreted as the marginal change in the probability of survival of a southern firm compared to the control group in month t . For months prior to the reform, β_S^t has a slightly different interpretation, identifying the marginal change in the probability of being operating in the market compared to the control group. The analogous interpretation applies for the north group. By construction, $D_{i,t} = 1 \forall t \in (2014 - 2017)$, so those months will be omitted in the estimation.

The identifying assumption for Equation 3 is that, absent the tax change, exit rates would have evolved similarly across the southern, northern, and control groups. This assumption is plausible because the sample is the balanced set of firms active during the short-run window — a group of stations that were operating under comparable market conditions at the time of the reform, differentiated only by location relative to the border. The estimation strategy focuses on the extensive margin for the cohort of firms operating at the time of the shock. New entry after the reform is governed by location rules that depend on post-shock conditions and are outside the scope of this paper.

The results are presented in Figure A.4. The red and blue lines represent the associated coefficient for each period for the north and southern group, respectively. There are two vertical lines that represent the short run analysis period during which the estimation of the equation is omitted. In the short run, given the (literally) sunk costs, it is difficult to think reallocation of firms will take place. That is the reason why Section 3.1 used the time span of 2014 -2017 and here is used as the base period. The blue line shows the estimated coefficients for the southern group. The probability of survival is 5 percentage points higher in the southern group on average, increasing to almost 8 percentage points seven years after the policy change. The red line in Figure A.4 shows the coefficients for the northern stations. Two years after the policy, the survival rate decreases compared to both the control and southern group¹⁶.

Figure A.25 provides supporting evidence for the identifying assumption: pre-reform coefficients for both groups are close to zero and statistically indistinguishable from the control group, consistent with parallel pre-trends in survival rates. Figure A.20 corroborates this by showing a similar evolution in the total number of firms across regions before the reform.

These results bridge two branches of the literature analyzing CBS. On the one hand,

¹⁶It is worth noting that the difference in the survival rate for the northern group gets closer to zero towards the end of the period, seven years after the tax change. A possible interpretation is this the time it took to adjust to the new long run equilibrium.

papers analyzing short-run results exploit shocks in CBS incentives taking location decisions as given¹⁷. On the other hand, papers that characterize long-run equilibria abstract from shocks to CBS incentives (Chiou and Muehlegger, 2008; Manuszak and Moul, 2009). The panel structure of the data allows me to analyze how firms shocked negatively in the short run are the ones that increase their probability of exit in the long run. A key requirement for detecting firm exit is a shock that is both large and persistent. The quantity responses documented in Section 3 show a sharp, immediate divergence across the border, and Figure A.19 confirms that the price differential remained substantial over the following seven years despite subsequent tax adjustments¹⁸.

Two related studies link CBS conditions to firm survival. Campbell and Lapham (2004) find that exchange rate differentials increase aggregate store counts in US border counties facing Canada. Baggs et al. (2016) find no significant effect of exchange rates on firm survival at the Canada-US border, though they do find effects on employment and sales. This paper complements their results by using tax variation rather than exchange rates, by exploiting direct firm-level quantity data, and by identifying both positive and negative survival effects depending on which side of the border a firm is located on. More broadly, the results speak to the vast literature examining how state tax differentials affect business location at geographical borders (Bartik, 1985; Holmes, 1998; Peltzman, 2016; Rohlin et al., 2014), providing a context where firm exit is induced by tax differentials via CBS.

5 Discussion: Reconciling Prices, Quantities, and Exit

The evidence in Sections 3 and 4 produces five facts. *Fact 1*: the federal tax cut is incompletely passed to retail prices south of the border. *Fact 2*: retail prices north of the border did not fall (even increased for some specifications), despite southern stations becoming cheaper competitors. *Fact 3*: On both sides, price responses is homogenous within region. *Fact 4*: quantities sold at southern stations near the border increased substantially, while northern border stations lost sales persistently. *Fact 5*: the probability of survival diverged sharply across the border in the long run, with northern stations near the boundary closing at substantially higher rates. These five facts are, at first glance, difficult to square with the standard prediction that CBS should generate price convergence at the border. The starting point of this section is to ask whether price convergence is a necessary consequence of CBS, or whether there exist market structures and institutional settings in which CBS and spatially uniform pass-through can coexist. The answer to this question will help identify which models are consistent with the results found in the previous sections and rule out those that are not.

¹⁷Bajo-Buenestado and Borrella-Mas, 2019; Burstein et al., 2024; Coyne, 2017; Doyle and Samphantharak, 2008; Gehrsitz et al., 2021; Harding et al., 2012; Hindriks and Serse, 2019; Jansen and Jonker, 2018; Louis and Taylor, 2016; Messner et al., 2024; Teixidó et al., 2024; Tsvetanov, 2024.

¹⁸In March 2018 the ad-valorem tax was substituted by a per-unit tax, updated several times for inflation.

In order to explore the different explanations, I will first present the standard conceptual framework that will predict the heterogeneous pass-through with respect to the distance to the border. Assume there are two markets in different regions within a state, a *central* one, and a *border* one close to another state with the same standard demand $D(p)$ and supply $S(p)$ for gas. Assume a tax τ is levied on both regions. CBS can be formalized as the border region having a more elastic demand¹⁹, given that consumers can substitute consumption and go to the other side of the state to purchase the same good. The equilibrium result will be the one depicted in Figure A.5, where the pass-through of the tax will be higher in the central region compared to the border region ($\Delta P_c > \Delta P_b$). On the other state, the opposite effect will take place in the border region due to an increase in demand, generating an increase in prices, known as *cross-border pass-through* (Bajo-Buenestado and Borrella-Mas, 2019). The heterogeneous pass-through within each region will then be a sufficient test to detect CBS (Bajo-Buenestado and Borrella-Mas, 2019; Coyne, 2017; Doyle and Samphantharak, 2008).

However, Fact 3 shows a lack of a heterogeneous response in prices on each side of the border. The literature that finds a lack of price convergence often associates it with the absence of CBS. That is, either markets are not well integrated and the costs of traveling are too high (Louis and Taylor, 2016), or consumers and firms are inattentive or uninformed about the tax change and do not internalize the price differential (Hindriks and Serse, 2019). It could also be the case that historically the price differential had been so sharp that firms had already exited the market and the new tax change did not make any difference (Jansen and Jonker, 2018). In terms of Figure A.5, it would simply mean that in the border region, the $D(p)$ does not change and behaves as in the central region.

Even though the lack of CBS can account for the lack of price convergence, the response in quantities argues in favor of significant CBS among regions. The literature has suggested other reasons that can explain a situation with no price convergence. As mentioned by Doyle and Samphantharak (2008), in a perfectly competitive market, heterogeneous pass-through will not emerge if the supply exhibits a constant marginal cost. Figure A.6 shows the case of a horizontal $S(p)$, where pass-through is identical across regions despite cross-border shopping.

The assumption of perfect competition with constant marginal costs might be too restrictive and is not the only potential explanation for the lack of price convergence. In a perfectly competitive market, the pass-through of a tax is a function of demand and supply elasticities. In non-competitive markets, pass-through will depend on the convexity of the demand and the degree of competition (Weyl and Fabinger, 2013). Figure A.7 presents a case with a monopoly and a more elastic and convex demand in the border. The equilibrium price after the tax is the same in the central p_1^c and border p_1^b regions after the tax, even though the quantity decreases more in the border region.

¹⁹A similar thought experiment could be done, but instead of a change in the elasticity, an inner shift of the demand in the border region as in Bajo-Buenestado and Borrella-Mas, 2019.

Another potential story would be related to the flexibility of firms in adjusting prices. Assume there is a market friction where prices are rigid. The former could be induced by the government as a form of maximum price, as is common in energy markets (Guenette, 2020; Kojima, 2013). Price rigidity could also arise from multi-establishment firms playing a strategy of homogeneous pricing (Friberg et al., 2022; Messner et al., 2024). The result of this type of friction would still be a homogeneous pass-through of taxes and presence of CBS, as depicted in Figure A.8.

The purpose of this section is to present a discussion of potential settings where the price convergence test would fail in the presence of CBS. Moreover, CBS might be amplified or reduced by different market structures and institutional settings. Given the five facts stated before, there are some channels that can be ruled out in this setting. It would be difficult to argue that the reason for homogeneous pass-through is that there is a situation with constant marginal costs and perfect competition. The market is highly integrated, and there is a small set of firms that refine and distribute gasoline²⁰. Even assuming the refining industry is competitive, the assumption of perfect supply elasticity is hard to justify. It might be true for an individual firm that might have a constant marginal cost of purchasing a liter of gas up to the point the tank in the station is full, but it is more difficult to assume that this holds for the region itself. A story of full rigidity because of government intervention is also an incomplete explanation. Even though the government plays a role in the wholesale market (Kojima, 2013; Scheimberg, 2018), the pass-through in the south is less than 100%, which cannot be explained by the story of price rigidity alone. Section 6 finds no heterogeneity for different brands. YPF, the largest retailer and state-owned company, does not exhibit a different pattern in terms of pass-through, as depicted in Table A.12.

A potential mechanism that would reconcile the results is a story of price discrimination where firms with market power face two types of customers: less elastic *local consumers* and *cross-border shoppers* with negligible transport cost, who buy by comparing prices of many firms in different regions. A tax cut of the magnitude of 30% lowers prices in the south and induces customers to switch to the southern region, leaving northern stations facing the inelastic part of the demand, the immobile shoppers, who in principle could face increased margins given that firms now face the inelastic segment of the demand (Hindriks and Serse, 2019). Like the constant-marginal-cost case, this mechanism delivers Fact 1-4 but does not, on its own, predict the differential exit pattern of Fact 5. The story is completed by the fact that stations face a significant fixed cost of operating in the market. Even though margins might increase, the fall in quantity is so important that firms in the north will not recover the fixed cost and in the long run will tend to exit the market. Section .2 presents a formalization of these mechanisms where firms have to decide serving between two markets and recover fixed costs to sustain non negative-profit condition.

A corollary of this section is that using only prices to infer and reject CBS must be

²⁰Coloma, 2003; Coloma, 1998, 1999, 2002; Culós et al., 2022; Scheimberg, 2018; Serebrisky, 2001.

accompanied by an assumption about how the market works and its organizational structure. This lesson does not only apply to CBS in particular but to a broader set of questions related to taxation in general. Several papers test the heterogeneous pass-through of a tax along a dimension where firms differ to gain insights into the existence of a related phenomenon: the number of competitors to infer market power (Genakos and Pagliero, 2022), the identity of the tax remitter to infer monitoring costs and tax evasion (Kopczuk et al., 2016), the type of stores where anti-profiteering measures can be implemented to estimate redistributive effects of taxes (Benzarti et al., 2024), or regional wealth where firms are located to estimate the progressivity of gas taxes (Stolper, 2024).

6 Robustness

This section complements the short-run results in four dimensions to explore the nature of the results. First, because of the market structure and differences in brand concentration, I investigate if there are heterogeneous effects by brand that explain the results. Second, I assess whether the number of competitors around each gas station can reflect differences in pass-through (Genakos and Pagliero, 2022). Third, I analyze whether there exist anticipation effects or lags in the implementation of the price adjustment by exploiting a dynamic analysis. Fourth, I employ several definitions of treatment groups, to measure the distance at which one can find effects of CBS.

6.1 Heterogeneous effects per brand

As explained by Section 2, there are differences in brand concentration with YPF being a clear leader that accounts for more than 50% of market share. A natural question arises if the leader is setting a uniform price across the country, because then the average effect on prices will be mainly capturing only one firm’s decision. This uniform price strategy can arise in a context of government owned firm, or multi establishment firms that implement a strategy of uniform pricing (Friberg et al., 2022; Messner et al., 2024). In this section I differentiate the effects per brand on the evolution of price and quantity after the tax exemption to identify if smaller brands have any different strategy compared to the leader. The estimated equation is:

$$Y_{i,t} = \alpha + \sum_{k=1}^K \beta_S^t \times South_i \times I_k\{k\} \times Post_t + \sum_{k=1}^K \beta_N^t \times North_i \times I_k\{k\} \times Post_t + \eta_t + \psi_i + \epsilon_{i,t} \quad (4)$$

which is analogous to Equation 1 but uses an interacted variable for each brand k . Figure A.9 depicts the result for both types of gas. There is no significant differentiation between the different brands, although the point estimates for YPF are slightly higher compared to the others. Regardless of the former, Table A.12 shows the 95% confidence

interval for the pass-through for the different brands and the overlap between them. The quantity estimates are not different in terms of heterogeneity. There seems to be in the South a lower increase for regular gasoline, but is compensated with a higher increase in Premium for YPF. Overall, the data does not present a clear differentiation between coefficients for any particular brand.

6.2 Number of competitors

The nature of competition is a key dimension to determine pass-through in non perfectly competitive environments (Weyl and Fabinger, 2013). An implicit assumption of the results in the main section is that every firm faces the same intensity of competition. Genakos and Pagliero (2022) finds evidence of a higher pass-through in gasoline stations facing less competitors. This section tests for heterogeneous evolution of price and quantity based on the number of nearby competitors around each gas station. The estimated equation is

$$Y_{i,t} = \alpha + \sum_{c=0}^{3+} \beta_S^c \times South_i \times I_c\{c\} + \sum_{c=0}^{3+} \beta_N^c \times North_i \times I_c\{c\} + \eta_t + \psi_i + \epsilon_{i,t} \quad (5)$$

which is analogous to Equation 1 but uses an interaction based on the number of competitors around a certain radius. The number of competitors are defined using the three different variables of distances, euclidean, driving distance and driving time, for different radius, 2 and 5 km or minutes, respectively. Figures A.10 depict the estimated coefficient for different types of distance measures to define the number of competitors²¹. Overall, there is no clear pattern on heterogeneous effects that arise from the number of competitors in terms of pricing. Quantities present a similar story but the percentage fall in the north seems to be stronger for low competitive environments for regular gas. In general, the different definitions of distance to count the number of competitors do not present any major difference.

6.3 Dynamic effects

The methodology in Section 3.1 identifies the average post treatment effect assuming the policy had a homogeneous effect every month. In practice, this could not be the case. On the one hand, a concern would arise if anticipation effects of the policy are present (Coglianese et al., 2017). On the other hand, consumers and firms might learn over time from the policy change and adjust their decisions over different months. This section estimates the dynamic evolution of price and quantity after the tax exemption. The estimated equation is:

²¹See Regular gas in Figure A.22.

$$Y_{i,t} = \alpha + \sum_{t=nov15-L}^{nov15+F} \beta_S^t \times South_i \times I_t\{t\} + \sum_{t=nov15-L}^{nov15+F} \beta_N^t \times North_i \times I_t\{t\} + \eta_t + \psi_i + \epsilon_{i,t} \quad (6)$$

which is analogous to Equation 1 but uses an interacted binary variable for each month. L and F are the respective lags and leads for before and after the period (January-2014 and December-2017). The base period is November 2015, one month before the reform. Figure A.23 depicts the evolution for premium gas of gas²².

The pre-trends for prices are similar for both groups and around zero, meaning no significant difference with the control group before the reform and suggesting no presence of anticipatory effects. After the tax change, the effect on prices is immediate and permanent for both types of gas.

The effects on quantities are more nuanced. After the tax change, the quantity response is immediate and persists over time. Notably, the southern group presents evidence of an increasing trend over the post-treatment period, consistent with a gradual learning process among consumers that continues well beyond the month of the reform. The magnitude of this trend is, however, relatively small compared to the initial jump observed in the first month after the policy change. The same pattern holds for premium gas in the South.

For the case of northern stations, quantities are not distinguishable from zero in most months, even though the point estimates are negative. The analysis by type of gas is different for this group of stations. Premium gas seems to drop in the first months to then recover, where regular gas seems more stable over time.

6.4 Changing the definition of treatment groups

This section analyzes the sensitivity to the definition of the extension of the area around the border. In the baseline estimation, the exempted and non-exempted groups were defined as gas stations around 175 km (≈ 108.7 miles) south and north of the border. Figure A.12 shows the result of estimating Equation 1 using different extensions for defining the treatment groups. The results in terms of prices are not sensitive to the definition of the area around the border. The effects of the baseline are preserved: a sharp discontinuity in the price for the southern region and a slight increase in prices for the northern group.

In terms of quantities, the main conclusion from the baseline analysis is not changed. The effects are stronger closer to the border and attenuate as we incorporate stations located further with a difference in the regions. The coefficients of the northern region stabilize around zero as the region definition is expanded, consistent with the aggregation bias. The effects stop being significant around 200 km (≈ 133 miles). It is interesting to note that the former distance is really similar to the bin that was non significant in Section 3.3. On the other hand, the effects in the south do not converge to zero as the definition

²²See regular gas in Figure A.24.

is expanded. As discussed in Section 3.1, the existence of the older border truncates the number of firms that can be added to the group.

7 Conclusion

This paper studies the limits of relying on price data to make inference about cross-border shopping (CBS) behavior, a phenomenon in which consumers travel to neighboring regions to take advantage of lower prices. I exploit a large and exogenous reform to Argentina’s federal gasoline excise tax, enacted in December 2015, which expanded a pre-existing tax-exempt zone in southern Argentina and reduced the applicable rate from 55% to 5%—a roughly 30% drop in the tax-inclusive retail price. The reform was politically motivated rather than driven by contemporaneous market conditions, took effect within days of being announced, and generated a sharp geographic discontinuity in the tax burden faced by otherwise comparable gasoline stations on either side of the border. To study the response of consumers and firms, I build a novel station-level monthly panel covering the universe of Argentine gas stations, which jointly observes retail prices, quantities sold, geographic coordinates, and brand affiliation. I employ a difference-in-differences approach to compare stations on each side of the border to a control group located far away from the border, assumed not to be affected by the policy.

The results document strong evidence of CBS between regions after the policy was implemented. In the short run, exempted stations south of the new border cut prices by approximately 24% and saw quantities rise by 14%, while non-exempted stations just north of the border saw quantities fall by about 8%, with little to no adjustment in prices. Decomposing these effects by distance to the border, quantity responses are tightly concentrated near the border and attenuate with distance. In contrast, prices remain homogeneous within each region, with no evidence of price convergence. In the long run, this short-run reallocation of demand translates into a divergence in firm survival rates: five years after the reform, the probability of remaining active rises by roughly 5 percentage points for southern stations and falls by a similar amount for northern stations near the border.

The implications extend beyond the specific Argentine context. The findings show that prices alone can be an insufficient instrument for detecting CBS: several market structures—competitive markets with constant marginal costs, heterogeneous demand elasticities arising from mobile and immobile shoppers, or price rigidities induced by regulation or uniform-pricing strategies of multi-establishment firms—can generate uniform pass-through alongside significant cross-border reallocation. Detecting CBS therefore requires either quantity data or explicit assumptions about the underlying market structure. More broadly, these results speak to a growing empirical literature that uses heterogeneity in tax pass-through across firms, locations, or competitors to draw inferences about market power, tax evasion, regulatory frictions, or distributional incidence.

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Appendix

.1 Long run relocation

Table A.17 shows the comparison of the distribution of stations before and after 5 years after the reform for the period 2020-2021 for the three groups defined in the short run analysis. The exit rate is higher for the north group, where firms decrease by -0.4% . In contrast, the increase in the number of firms is 6.0% and 5.2% in the south and control region, respectively. The composition and distribution of brands does not change significantly. Even though the number of firms in absolute value is not much for each group, it does not seem the exit rate is driven by any type of brand in particular in the northern region.

.2 A Simple Model of CBS Shoppers and local monopoly

Consider a typical firm i faces two types of consumers: *Local consumers* with linear demand and *Cross-border shoppers* with zero transport cost, who buy from any firm whose price is at or below an exogenous threshold p^* . If firm i prices at or below p^* , it captures a share s_i of these shoppers. Demand for firm i is therefore

$$D_i(p_i) = A_i - b_i p_i + s_i \mathbb{I}(p_i \leq p^*). \quad (7)$$

Firm i has marginal cost $c_i + \tau_i$, where τ_i is a per-unit tax, and pays a fixed operating cost F_i each period it is active. Assume a distribution of F_i .

Profit is

$$\pi_i(p_i) = (p_i - c_i - \tau_i) [A_i - b_i p_i + s_i \mathbb{I}(p_i \leq p^*)] - F_i. \quad (8)$$

and the participation restriction is given by

$$\max\{\pi_i^H, \pi_i^B(p^*)\} \geq F_i, \quad (9)$$

where π_i^H and $\pi_i^B(p^*)$ in this expression denote operating profits (gross of F_i). The firm's best response is therefore the better of two local optima, with F_i entering only the entry/exit cutoff.

High-price strategy. Ignore cross-border shoppers and serve local demand only:

$$p_i^H = \frac{A_i + b_i(c_i + \tau_i)}{2b_i}, \quad \pi_i^H = \frac{(A_i - b_i(c_i + \tau_i))^2}{4b_i}. \quad (10)$$

Bunching at p^* . Match the threshold and capture the shoppers²³ :

$$\pi_i^B(p^*) = (p^* - c_i - \tau_i) (A_i - b_i p^* + s_i). \quad (11)$$

²³ Assume the relevant case where the unconstrained low-regime price $p_i^L = (A_i + s_i + b_i(c_i + \tau_i))/(2b_i)$ exceeds p^* , so that the bunching constraint binds.

The decision of the pricing strategy will depend on

$$s_i (p^* - c_i - \tau_i) \geq b_i (p_i^H - p^*)^2, \quad (12)$$

which expresses the trade-off transparently: the gain from capturing s_i shoppers at margin $p^* - c_i - \tau_i$ must outweigh the quadratic envelope cost of pricing below the unconstrained optimum. Substituting p_i^H , firm i prefers to bunch at p^* if

$$\Delta(\tau_i) \equiv (p^* - c_i - \tau_i) (A_i - b_i p^* + s_i) - \frac{(A_i - b_i(c_i + \tau_i))^2}{4b_i} \geq 0 \quad (13)$$

Taking derivative with respect to the tax

$$\frac{\partial \Delta}{\partial \tau_i} = -s_i - \frac{A_i + b_i(c_i + \tau_i) - 2b_i p^*}{2} < 0 \quad (14)$$

There exists a critical tax level $\bar{\tau}_i$ such that $\Delta(\bar{\tau}_i) = 0$. For $\tau_i > \bar{\tau}_i$, the firm abandons the CBS market and serves local consumers only at p_i^H .

The south's tax cut operates as an effective tax increase for northern firms through its effect on the competitive threshold p^* . When southern firms face a lower tax, they price lower, compressing the threshold that northern firms must match to retain cross-border shoppers. Formally, differentiating Δ with respect to p^* and substituting p_i^H ,

$$\frac{\partial \Delta}{\partial p^*} = 2b_i (p_i^H - p^*) + s_i > 0, \quad (15)$$

where the inequality follows from the bunching assumption $p_i^H > p^*$ and $s_i > 0$. A fall in p^* therefore reduces $\Delta(\tau_i)$ for every northern firm: the margin $(p^* - c_i - \tau_i)$ on cross-border shoppers shrinks, making the bunching strategy less attractive. Some northern firms that were previously bunching find $\Delta(\tau_i) < 0$ and optimally switch to the high-price strategy p_i^H .

This switch generates three consequences that map directly to the four empirical facts. First, firms that switch no longer capture cross-border shoppers, so their quantity falls while their posted price p_i^H is unchanged — consistent with Facts 2 and 3 simultaneously. The existence of fixed cost in the north will generate that firms whose best operating profit no longer covers F_i , the participation constraint is violated and they exit even without reducing margins.

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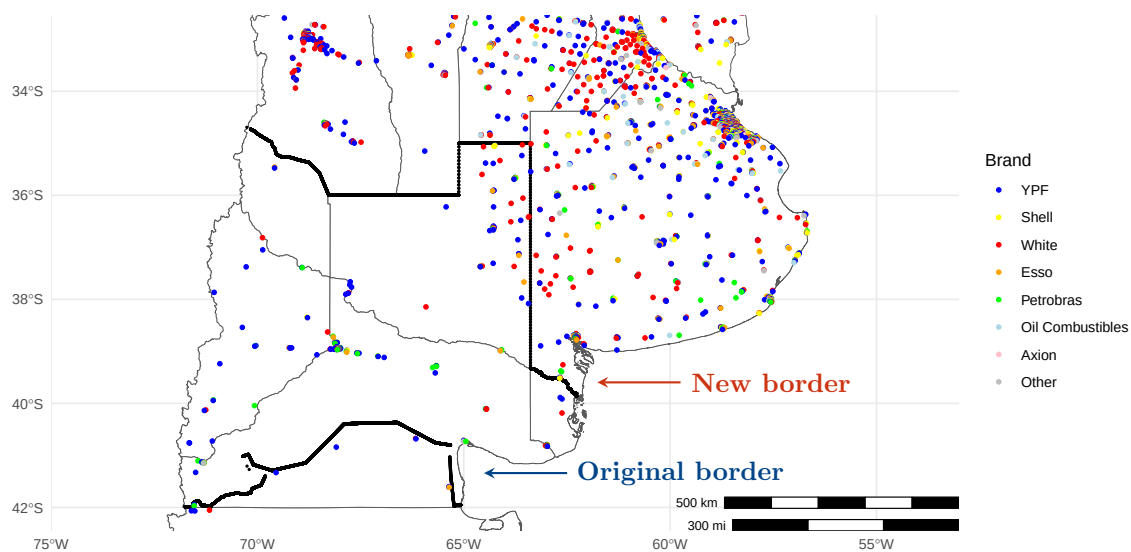
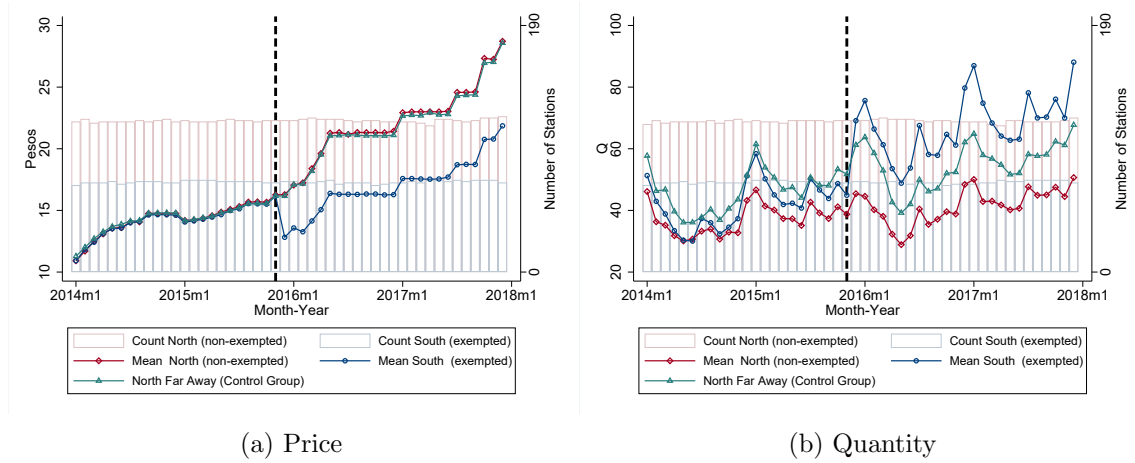


Figure A.1: The 2015 TTLF Tax Exemption Reform

Note: Dots represent the locations of gasoline stations. Colors indicate station brand. The blue line denotes the original TTLF exemption boundary (“42nd parallel zone”), while the red line denotes the expanded exemption boundary implemented in December 2015.



Note: This figure displays the monthly evolution of average retail prices (panel (a)) and quantities sold (panel (b)) for premium gasoline. The three series correspond to tax-exempt stations south of the new tax border (blue), non-exempted stations north of the border (red), and a northern control group located far from the border (green). Prices are in real pesos of December 2015; quantities are total station sales for premium gasoline in the administrative data. Both panels use the same group definitions: 175km ($\approx$ 109miles) north and south of the border. The control group consists of gas stations 550–750 km (342–466 miles) located north of the border. A vertical marker indicates December 2015, when the exemption took effect. The right axis shows the number of gas stations for each treatment group. Own elaboration based on Argentinian Ministry of Energy data.

Figure A.2: Price and quantity evolution around the border and control group, premium gasoline

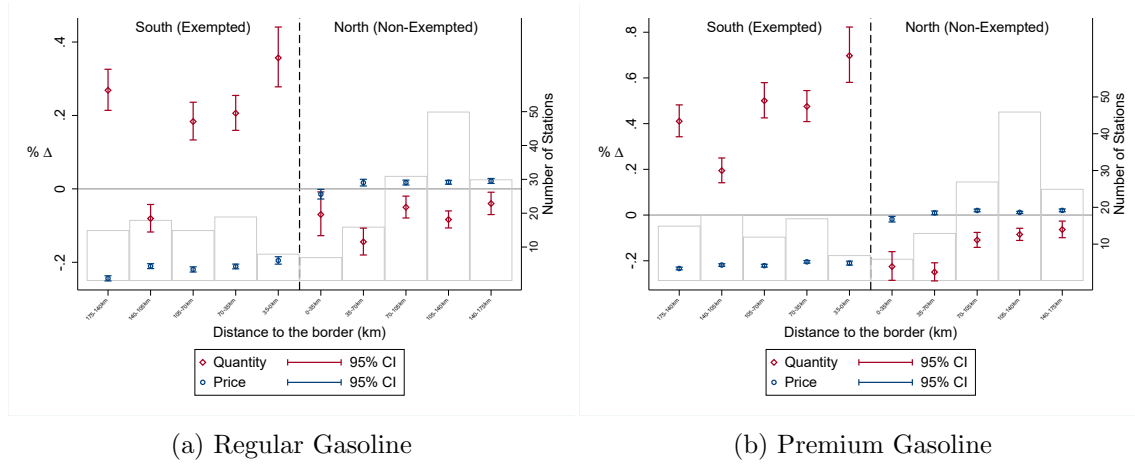


Figure A.3: Bins and estimates on price and quantities

Note: Each panel reports estimated post-reform effects by distance bin from Equation 2 for regular (left) and premium (right) gasoline. Blue markers are price effects (percent change relative to the control group); red markers are quantity effects. Bins are defined by driving distance to the closest station on the other side of the tax border; the control group is stations 550 – 750 km (342 – 466 miles) from the border (see text). Point estimates and confidence intervals correspond to the regression specification in Section 3.3. Tabulated coefficients appear in Tables A.8–A.9. Own elaboration based on Ministry of Energy data.

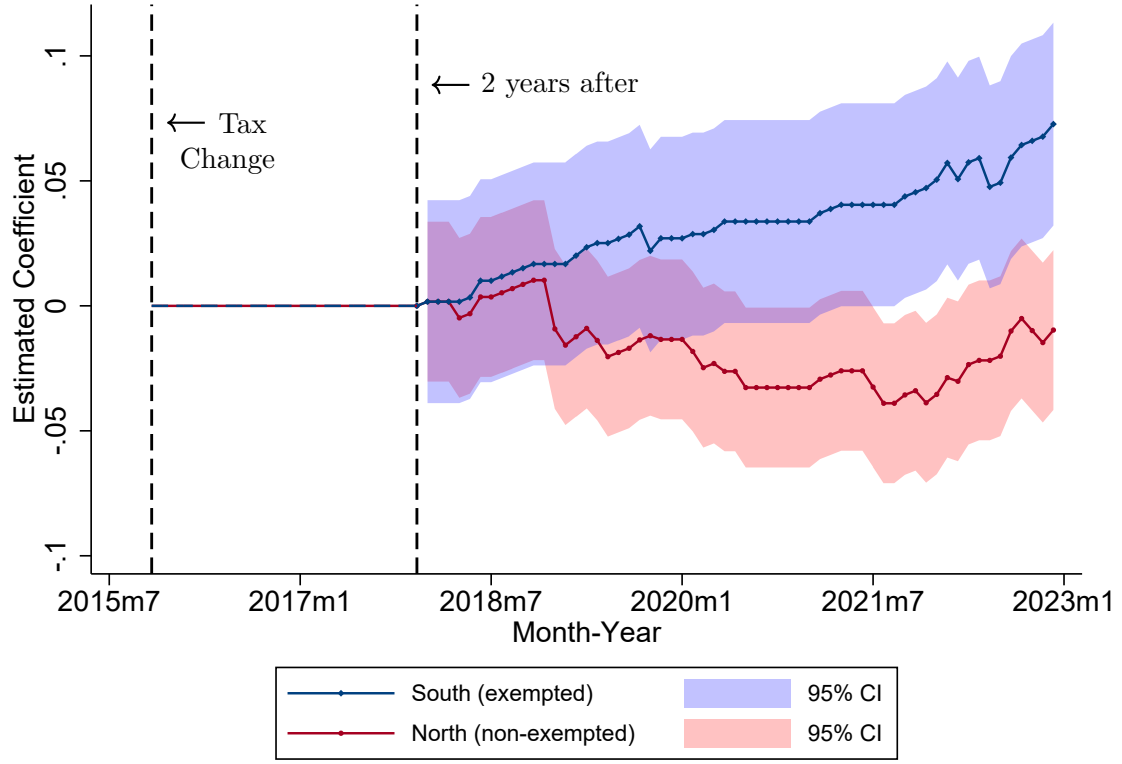


Figure A.4: Estimated coefficients for the probability of operating in the market

Note: The figure plots coefficient estimates from Equation 3 for the probability that station i is active in month t , relative to the northern control group. Blue (red) corresponds to the southern exempted (northern non-exempted) border sample. The balanced short-run window around the reform is omitted in estimation by construction (see text). Vertical markers indicate the reform date and the window labeled in the figure (two years before/after). Shaded bands are confidence intervals at 95% levels. Standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

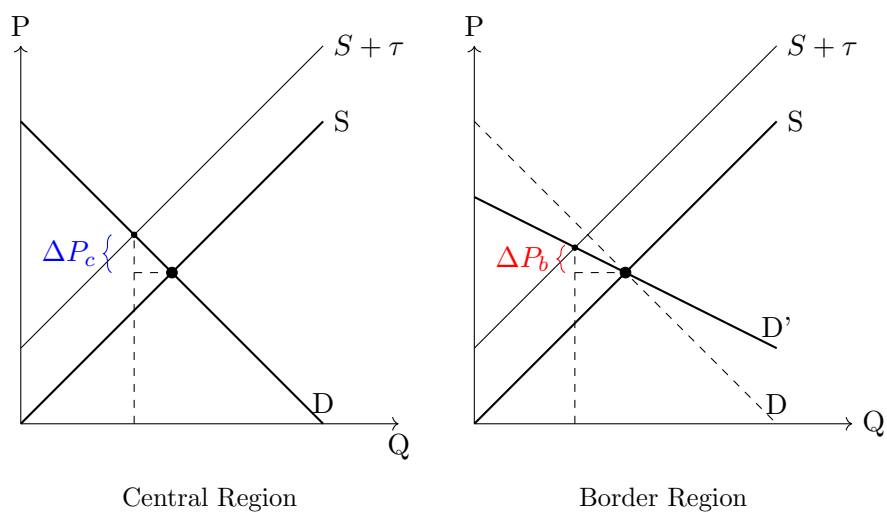


Figure A.5: Two regions and CBS

Note: Stylized supply–demand diagram for a central region (left) and a border region with outward-shifted demand D' capturing cross-border shopping (right). A specific excise τ shifts supply to $S + \tau$. ΔP_c and ΔP_b denote retail price pass-through in each region.

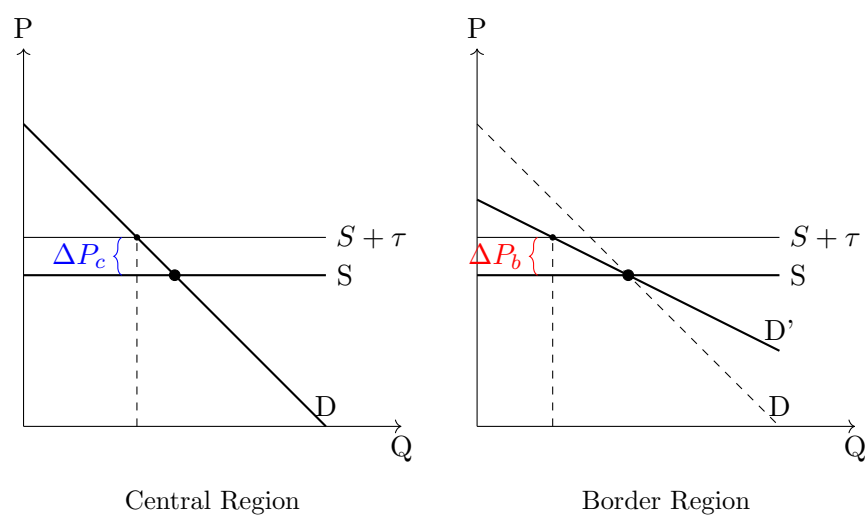


Figure A.6: Two regions and CBS with constant marginal costs

Note: Same labeling as Figure A.5, except marginal cost is perfectly elastic (flat supply), implying identical pass-through $\Delta P_c = \Delta P_b$ across regions despite cross-border shopping.

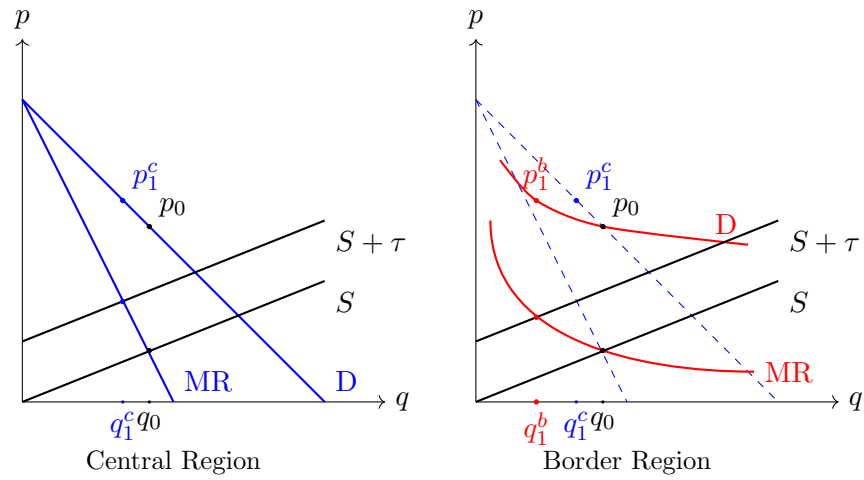


Figure A.7: Convexity and market power

Note: Monopoly pricing in a central region (linear demand) and a border market with a more elastic, convex demand curve. After the tax, equilibrium retail prices can coincide ($p_1^c = p_1^b$) while quantities fall more sharply in the border market.

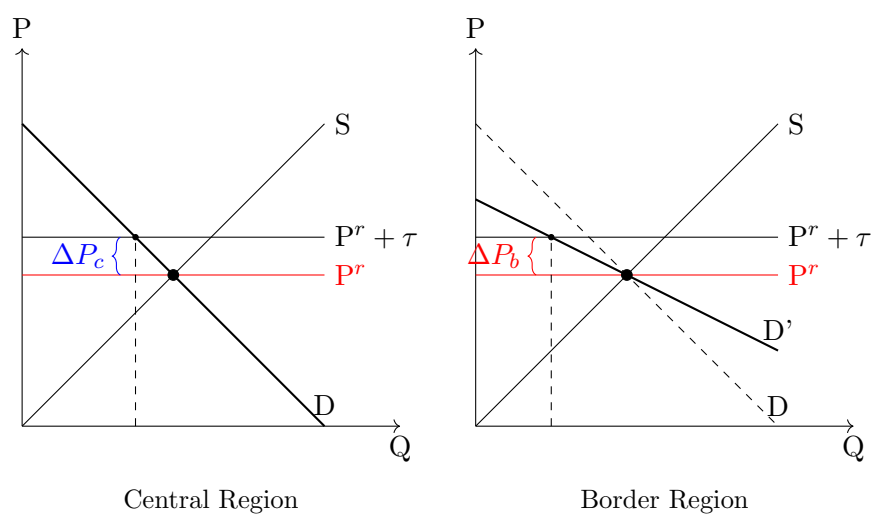


Figure A.8: CBS and price rigidity

Note: Illustration with a binding retail price cap P^r in both regions. The statutory shift to $P^r + \tau$ yields identical pass-through ($\Delta P_c = \Delta P_b$) even when border demand shifts to D' .

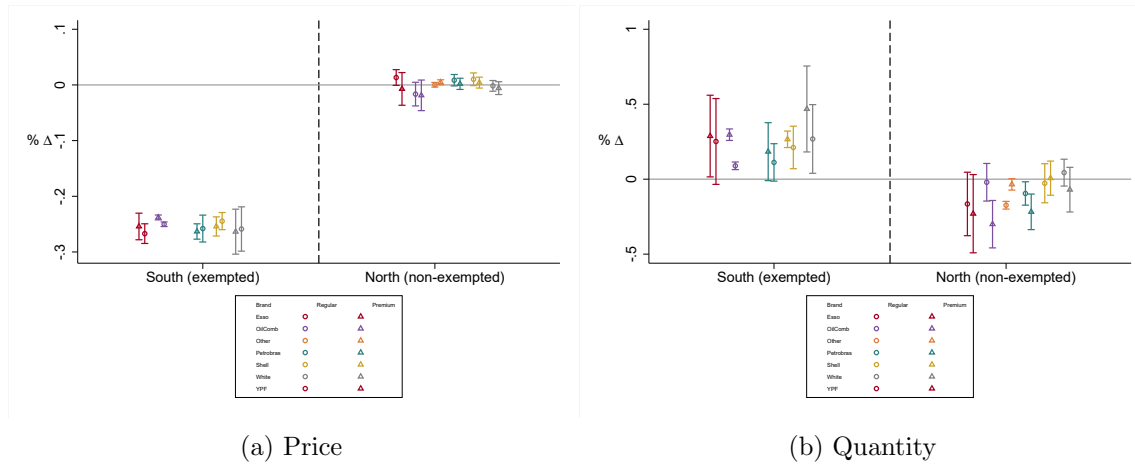


Figure A.9: Estimated coefficients for price and quantity differentiated by brand
Note: Point estimates from Equation 4 with log price (left) and log quantity (right), interacted with brand indicators for $South_i \times Post_t$ and $North_i \times Post_t$. The sample and fixed effects match the baseline short-run specification (station and month FE). Standard errors are clustered at the county level. Corresponding regressions are in Tables A.12–A.11. Own elaboration based on Ministry of Energy data.

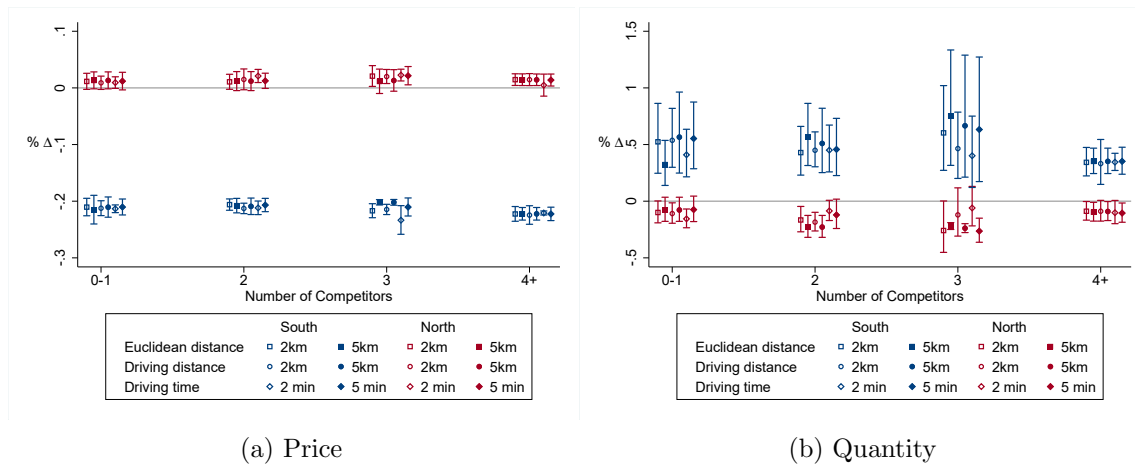


Figure A.10: Premium gas

Note: Estimated effects from Equation 5 for Premium gasoline, splitting treated stations by the number of competitors within a fixed radius (Euclidean distance, driving distance, or travel time, as indicated on the horizontal axis). Left: log price; right: log quantity. All regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

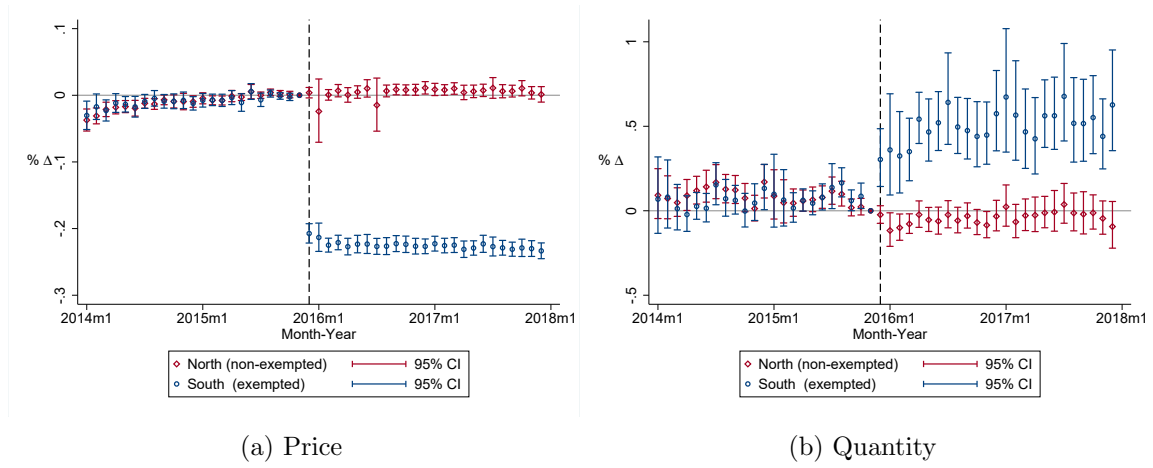


Figure A.11: Premium gas

Note: Dynamic difference-in-differences estimates from Equation 6 for Premium gasoline, with event months relative to November 2015 (omitted baseline). Coefficients are interactions of month dummies with $South_i$ and $North_i$; the control group is stations far from the border (see text). Left: log retail price; right: log quantity. Regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

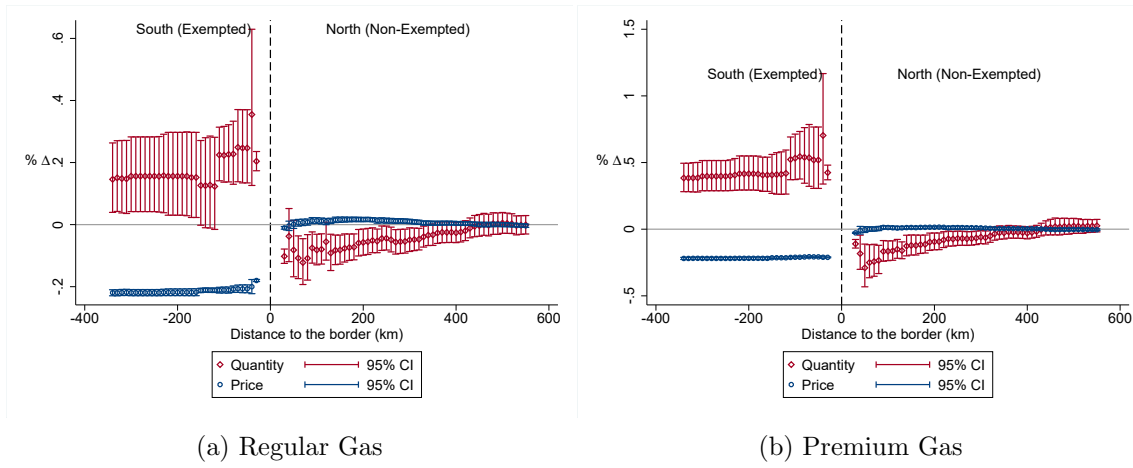


Figure A.12: Different extensions for defining each treatment group

Note: Sensitivity of estimates from Equation 1 to widening the treated bands north and south of the tax border. The horizontal axis indexes alternative distance cutoffs for inclusion in $South_i$ and $North_i$ (see text). All specifications include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

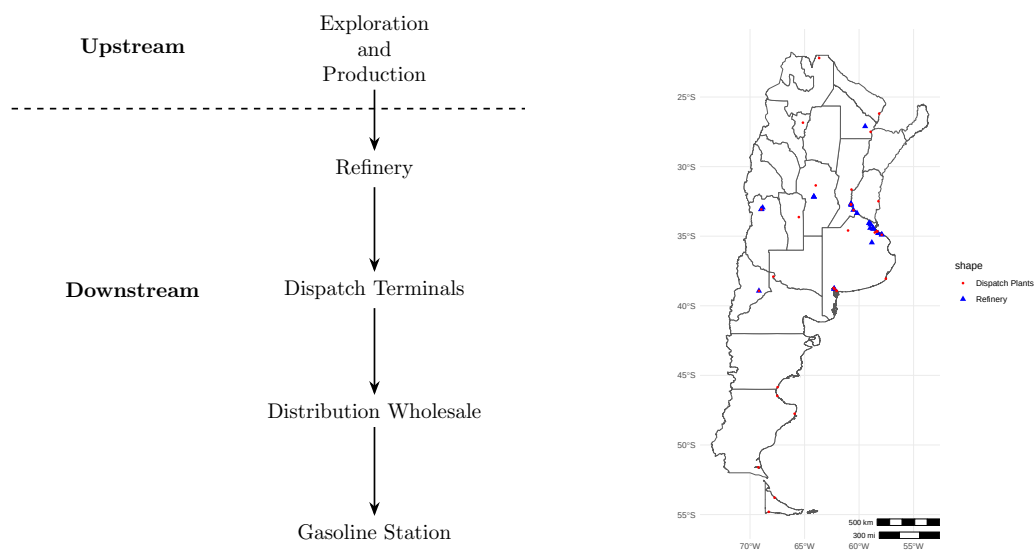


Figure A.13: Refineries and Dispatch Terminals

Note: Stylized schematic of the Argentine downstream chain from refineries to dispatch terminals and retail gasoline stations. Own elaboration for expositional purposes.

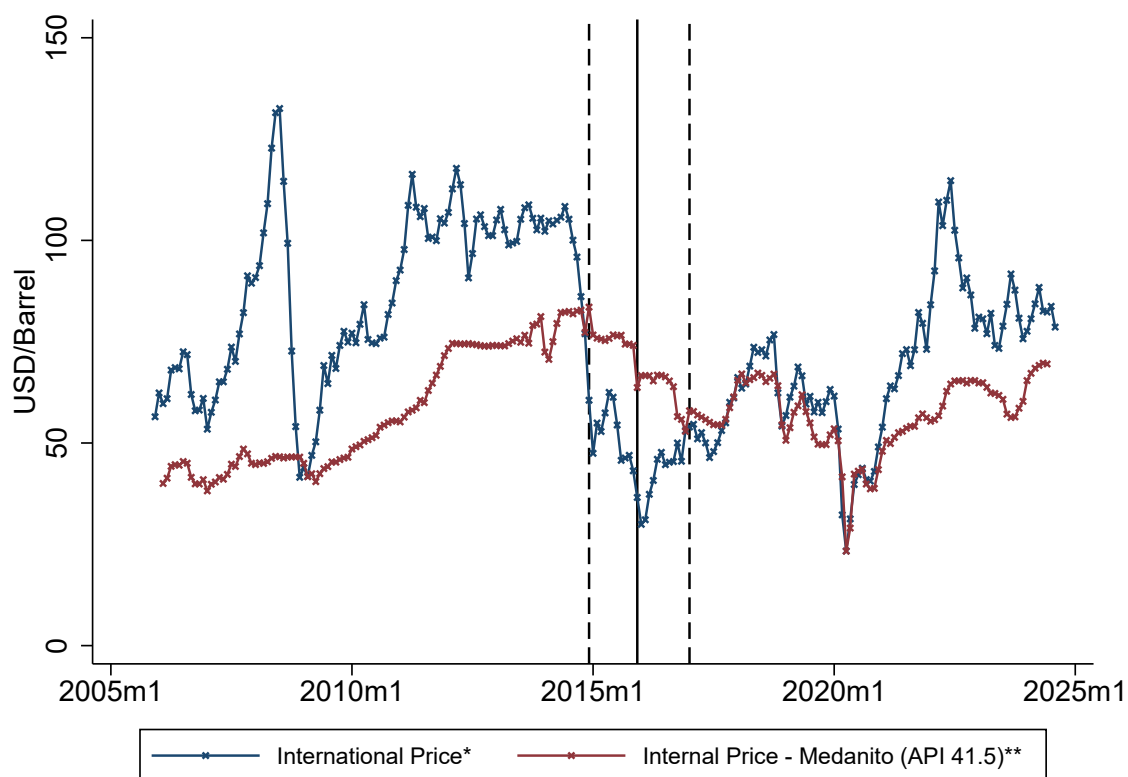
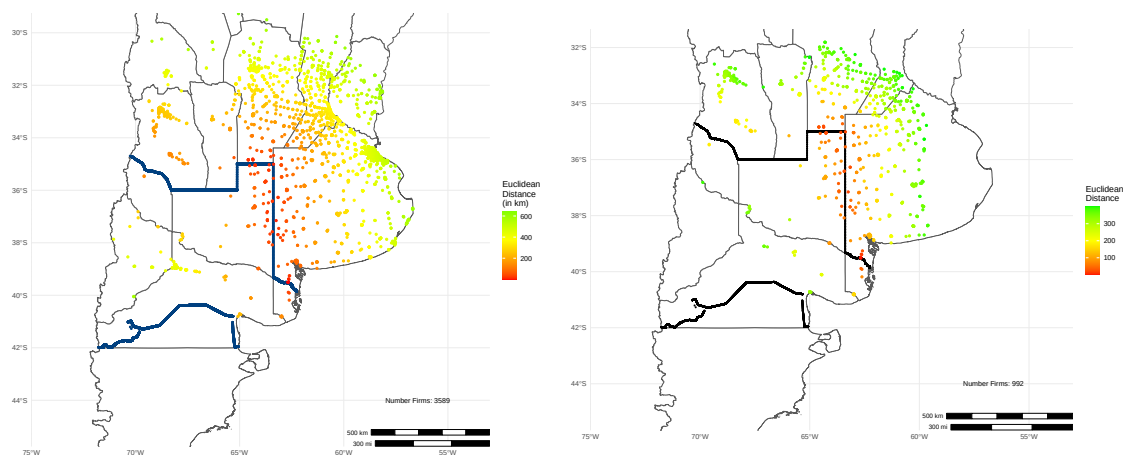


Figure A.14: Internal and International oil price

Note: See data sources below. * IMF commodity index (Average WTI,Brent and Dubai) ** Ministry of Energy of Argentina



(a) Euclidean Distance: **geodist** package

(b) Driving Distance: **osrm** package

Figure A.15: Distance to the closest station on the other side of the border.

Note: Maps plot the distance from each gasoline station to its nearest competitor located on the opposite side of the new TTLF exemption border. Panel (a) uses straight-line (Euclidean) distance computed with the **geodist** package Picard, 2010. Panel (b) uses route-based driving distance computed with the **osrmtime** package and OpenStreetMap OSRM Huber and Rust, 2016. Own elaboration based on Ministry of Energy data.

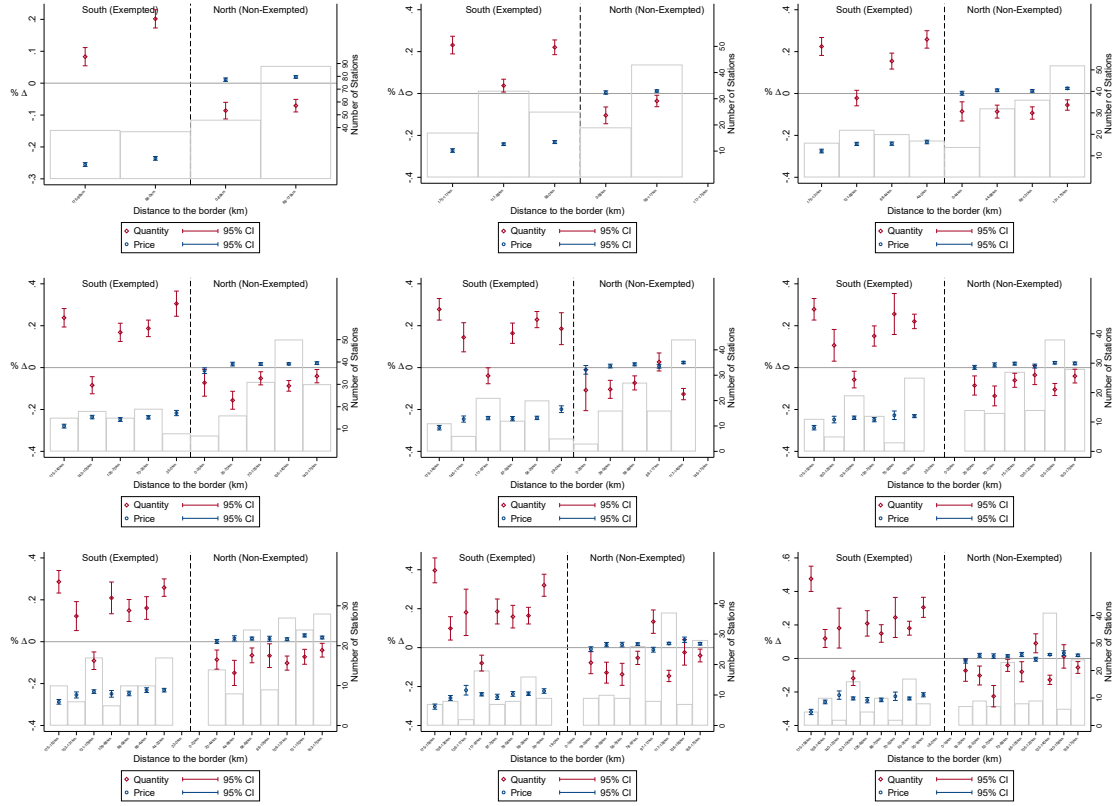


Figure A.16: Evolution of real price and quantities around the new tax border. Regular Gasoline

Note: Robustness to alternative bin widths for Equation 2: each small multiple fixes bin length to the value in the grid heading and replots price and quantity estimates. Group definitions: 175km (≈ 109 miles) north and south of the border; control group: 550 – 750 km (342 – 466 miles). Own elaboration based on Ministry of Energy data.

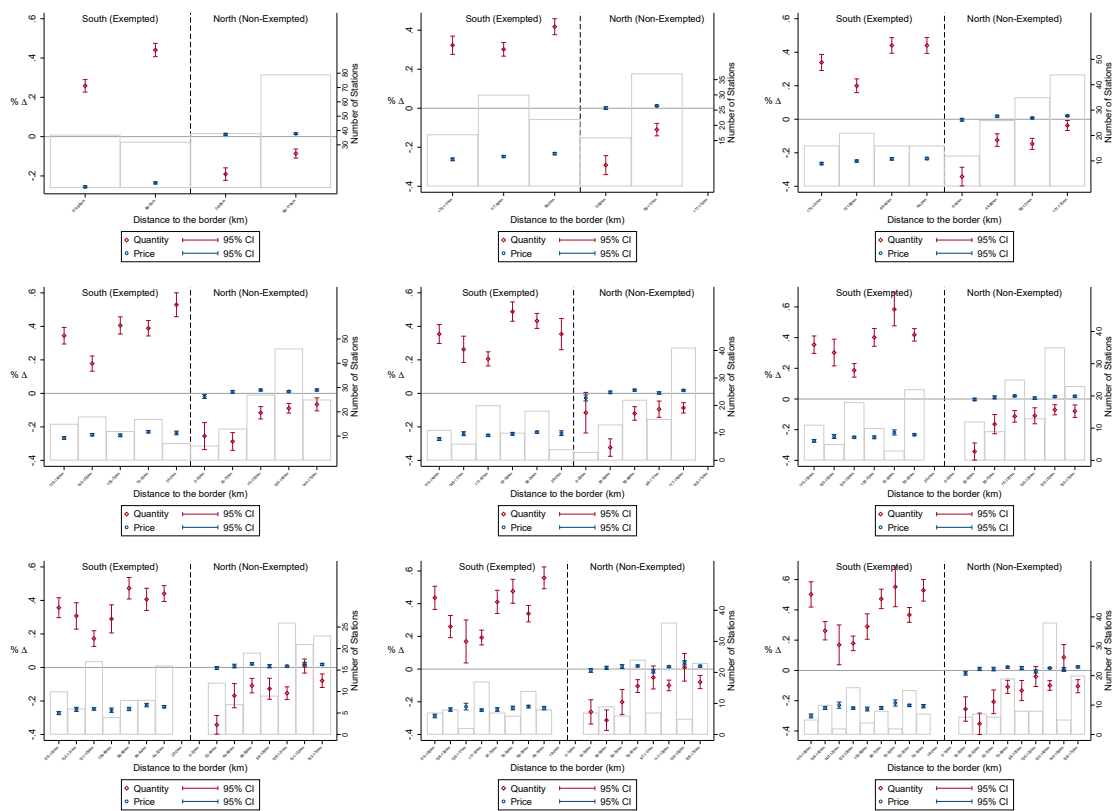
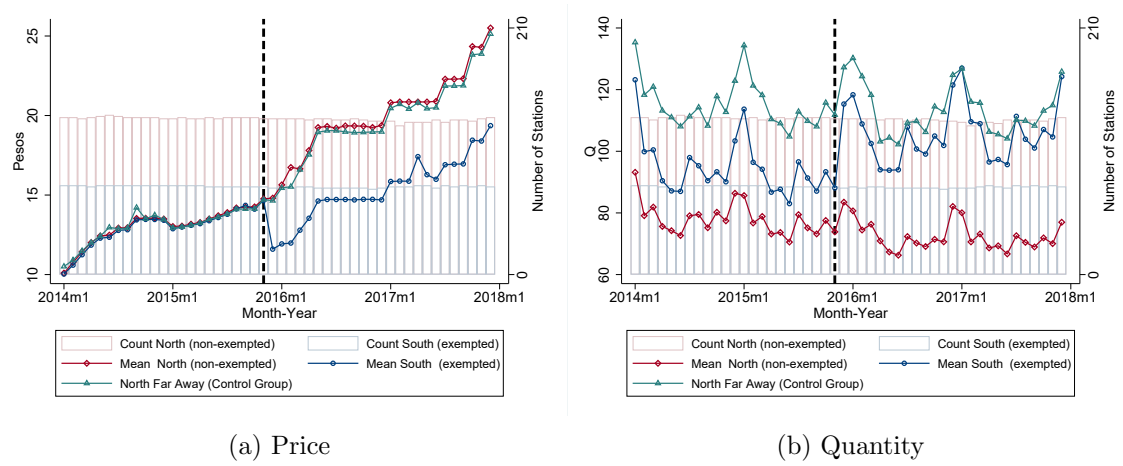


Figure A.17: Evolution of real price and quantities around the new tax border. Premium Gasoline

Note: Robustness to alternative bin widths for Equation 2: each small multiple fixes bin length to the value in the grid heading and replots price and quantity estimates. Group definitions: 175km ($\approx$ 109miles) north and south of the border; control group: 550 – 750 km (342 – 466 miles) . Own elaboration based on Ministry of Energy data.



Note: This figure displays the monthly evolution of average retail prices (panel (a)) and quantities sold (panel (b)) for regular gasoline for three groups: tax-exempt stations south of the new border (blue), non-exempted stations north of the border (red), and the northern control group (green). Group definitions: 175km (≈ 109 miles) north and south of the border; control: 550 – 750 km (342 – 466 miles). Vertical marker: December 2015 reform. Own elaboration based on Ministry of Energy data.

Figure A.18: Price and quantity evolution around the border and control group

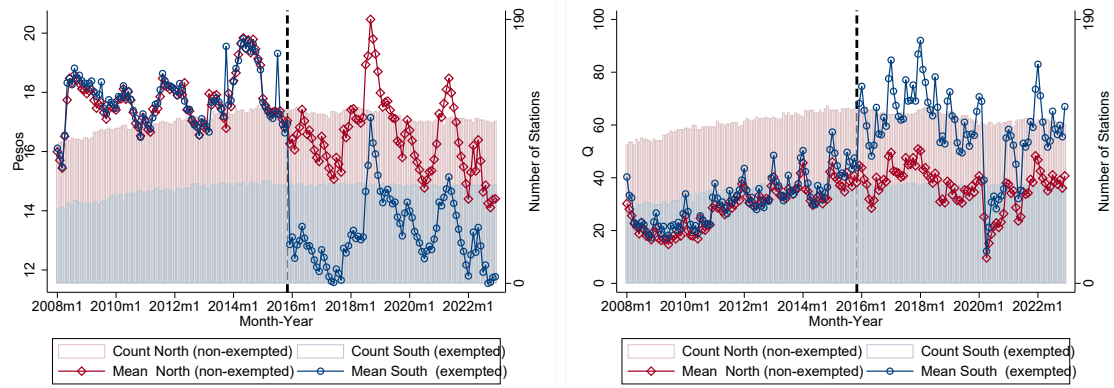


Figure A.19: Evolution of real price and quantities around the new tax border. Premium Gasoline

Note: Long-run evolution of average real retail prices (left) and quantities (right) for premium gasoline around the border. The plotted window and regional aggregation follow 175 km . Prices are in December 2015 pesos. Own elaboration based on Ministry of Energy data.

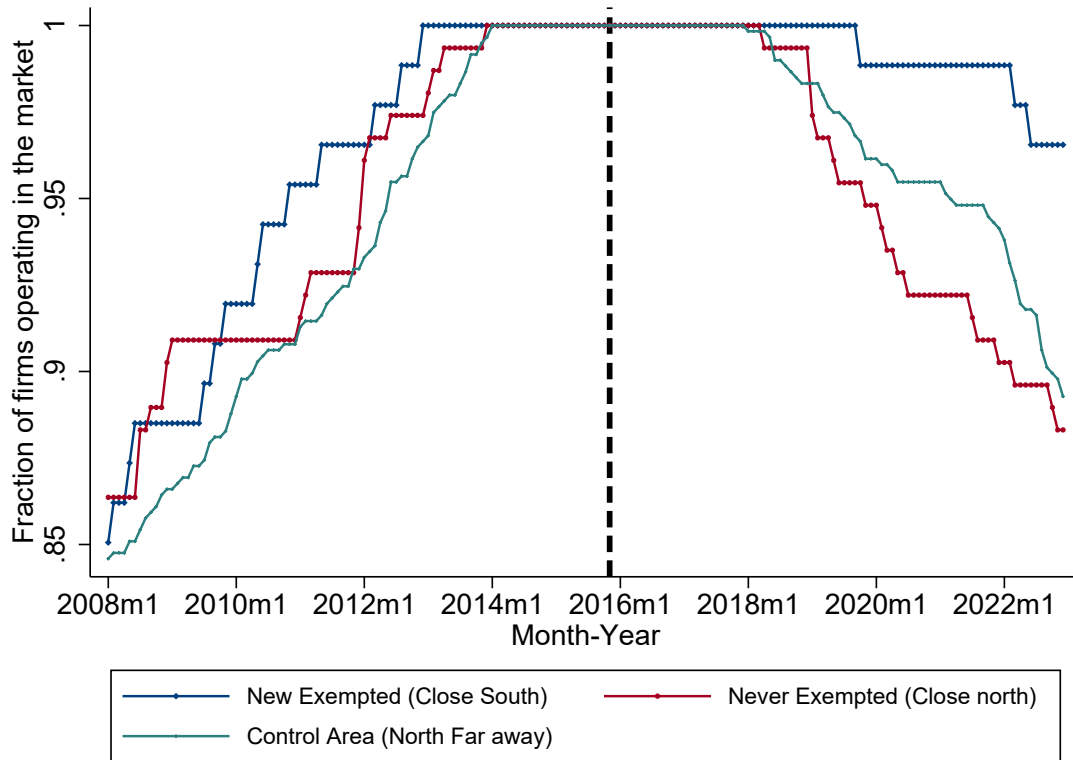


Figure A.20: Number of gasoline stations around the tax border.

Note: Count of active gasoline stations by month and region, normalized to 100 in the month immediately preceding the December 2015 reform. Regions: 175km ($\approx$ 109miles) north and south of the border; the control group (if shown) consists of stations 550 – 750 km (342 – 466 miles) . Own elaboration based on Ministry of Energy data.

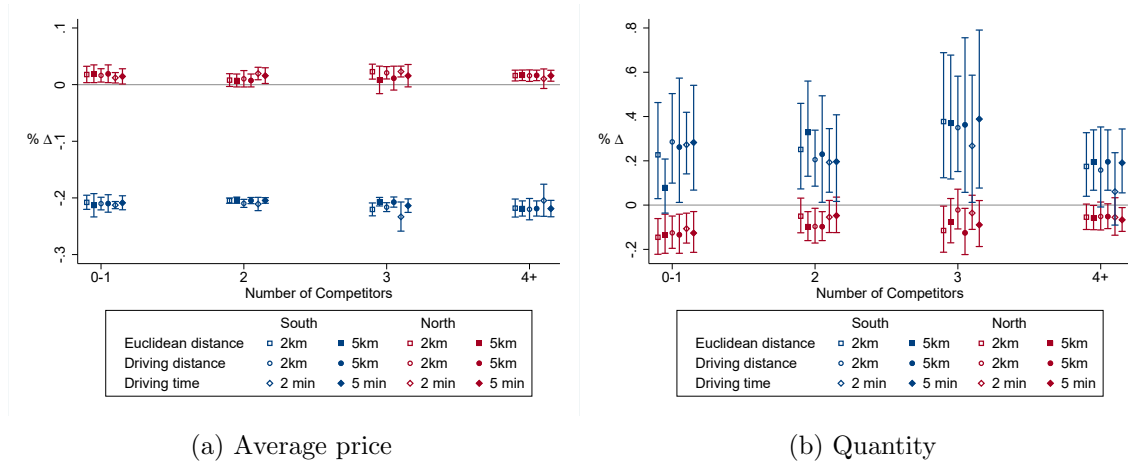


Figure A.21: Total gas (regular + premium)

Note: Estimated effects from Equation 5 for total gas (regular + premium), splitting treated stations by the number of competitors within a fixed radius (Euclidean distance, driving distance, or travel time, as indicated on the horizontal axis). Left: average log price; right: average log quantity. All regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

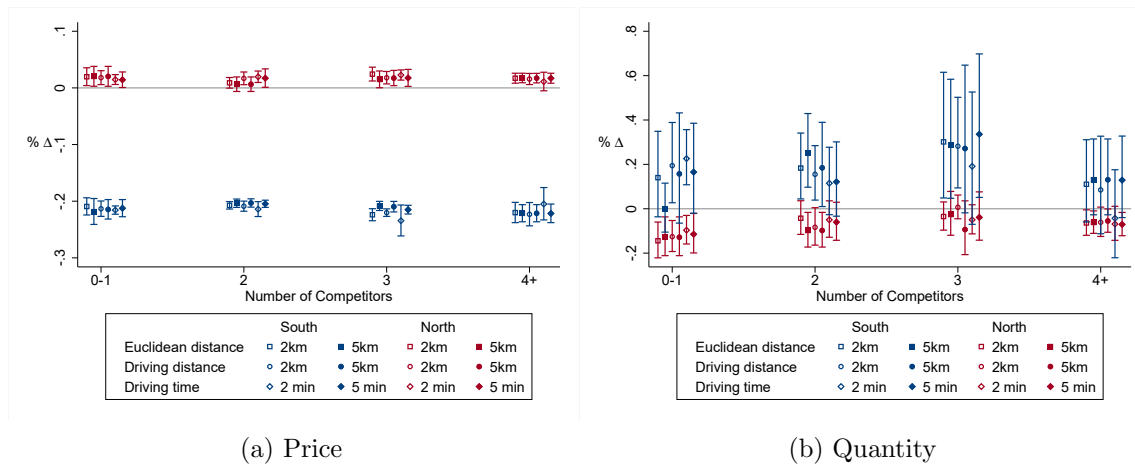


Figure A.22: Regular gas

Note: Estimated effects from Equation 5 for regular gasoline, splitting treated stations by the number of competitors within a fixed radius (Euclidean distance, driving distance, or travel time, as indicated on the horizontal axis). Left: log price; right: log quantity. All regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

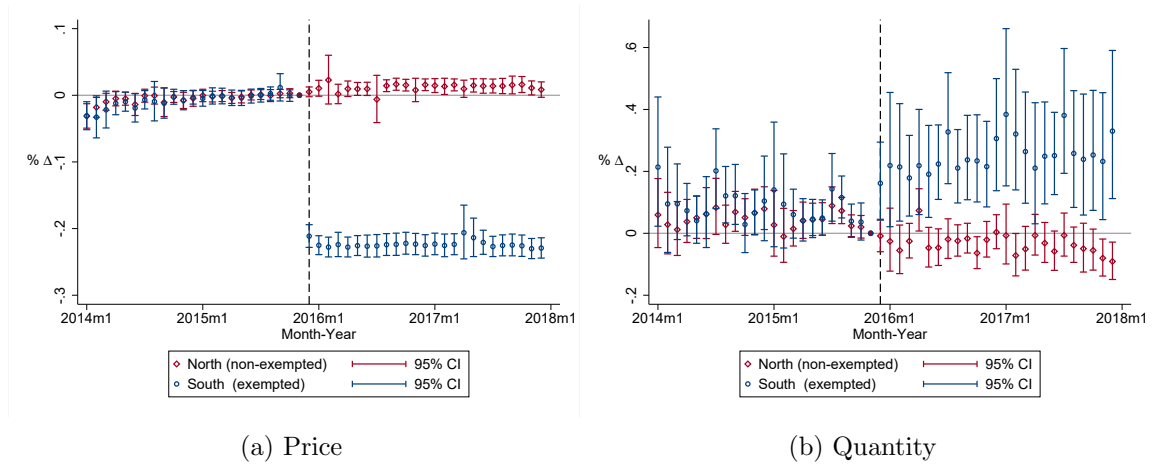


Figure A.23: Regular gas

Note: Dynamic difference-in-differences estimates from Equation 6 for regular gasoline, with event months relative to November 2015 (omitted baseline). Coefficients are interactions of month dummies with $South_i$ and $North_i$; the control group is stations far from the border (see text). Left: log retail price; right: log quantity. Regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

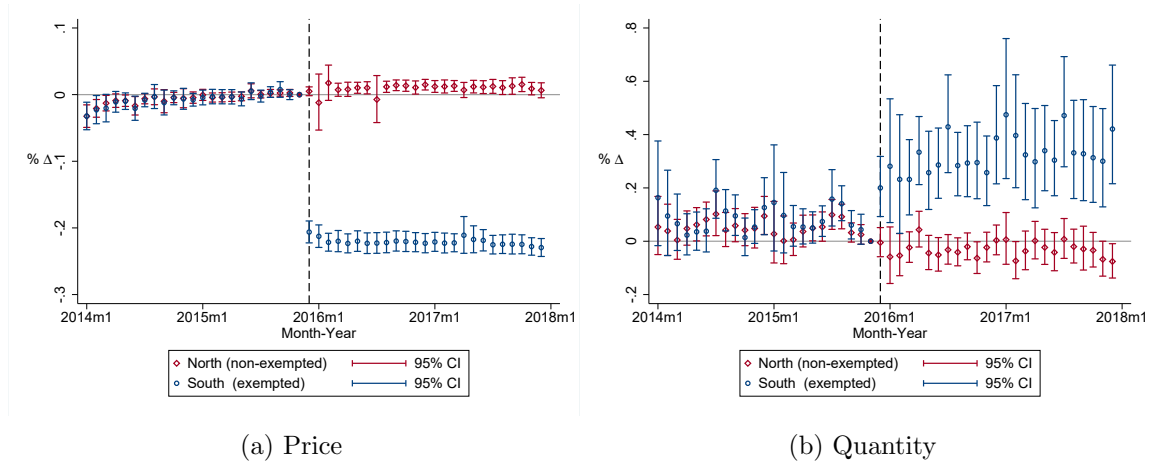


Figure A.24: Total gas (regular + premium)

Note: Dynamic difference-in-differences estimates from Equation 6 for total gas (regular + premium), with event months relative to November 2015 (omitted baseline). Coefficients are interactions of month dummies with $South_i$ and $North_i$; the control group is stations far from the border (see text). Left: average log retail price; right: log quantity. Regressions include station and month fixed effects; standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

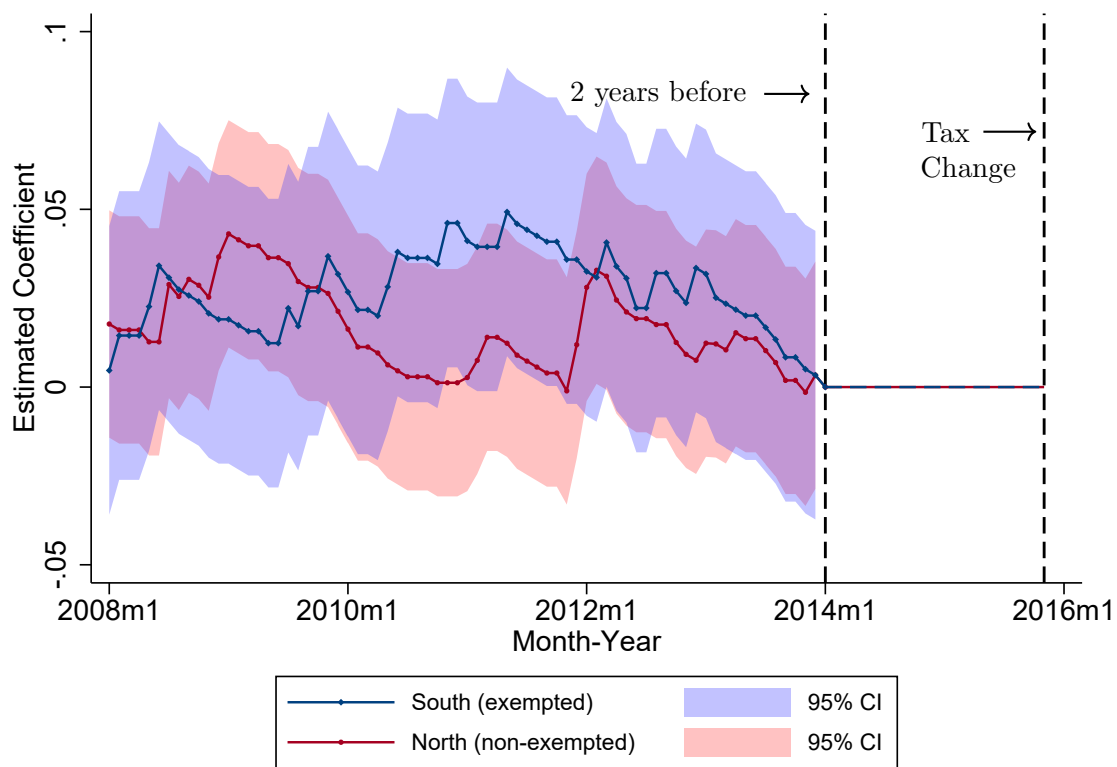


Figure A.25: Estimated coefficients for the probability of operating in the market

Note: The figure plots coefficient estimates from Equation 3 for the probability that station i is active in month t , relative to the northern control group. Blue (red) corresponds to the southern exempted (northern non-exempted) border sample. The balanced short-run window around the reform is omitted in estimation by construction (see text). Vertical markers indicate the reform date and the window labeled in the figure (two years before/after). Shaded bands are confidence intervals at 95% levels. Standard errors are clustered at the county level. Own elaboration based on Ministry of Energy data.

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Table A.1: Descriptive Statistics Complete Database

<i>Panel A: Price and quantity</i>						
VARIABLES	Regular Gas		Premium Gas			
	Mean	(SD)	Mean	(SD)		
Quantity (thousands liters)	117.5	(99.1)	48.6	(52.5)		
Price (pesos/liter)	14.8	(1.9)	16.6	(1.8)		
<i>Panel B: Distribution Stations and market shares</i>						
BRAND	N	%	% Regular	% Premium		
Axion	152	2.7	1.8	1.7		
Esso	484	8.6	8.9	6.4		
Oil Com.	313	5.6	3.8	2.0		
Other	330	5.9	2.9	1.0		
Petrob	334	6.0	5.6	3.8		
Shell	775	13.8	17.2	22.4		
White	1,438	25.7	5.1	2.5		
YPF	1,775	31.7	54.7	60.3		
Total	5,601	100.0	100.0	100.0		
<i>Panel C: Number of nearby stations around</i>						
DISTANCE	Radius	Mean	(SD)	Radius	Mean	(SD)
Euclidean	1 km	4.92	(3.94)	5 km	29.36	(36.70)
Driving	1 km	2.12	(1.26)	5 km	17.43	(19.51)
Time	1 min	1.46	(0.76)	5 min	9.58	(8.31)

Descriptive statistics for the whole universe of gas stations. Period: 2008 – 2022. Panel A shows total quantity in thousands of liters/gallons. Price is the average price reported by the gas stations each month. Prices are in pesos of December 2015 and converted into dollars using the exchange rate of December 2015 = 11.43. Quantities and prices are partialled out from seasonality patterns. Panel B shows the distribution of stations and markets shares for each brand for each type of gas. The number of stations is the total number of stations in the period. The market share is the average percentage of quantities per brand over the whole period. Panel C shows the average number of competitors around each gas station. The distance are presented in euclidean distance (using the *geodist* package—developed in Picard, 2010), and the driving distance (using the *osrmtime* package—developed in Huber and Rust, 2016—with the free Open Street Map OSRM software). 1 mile is equivalent to 1.6km. Own elaboration based on Argentinian Ministry of Energy dataset.

Table A.2: Descriptive Statistics before the reform

<i>Panel A: Price and quantity</i>							
VARIABLES		South		North		Control (550-750 km)	
		Mean	(SD)	Mean	(SD)	Mean	(SD)
Quantity (thousands liters)							
Regular Gas		95.84	(68.99)	79.26	(79.42)	116.92	(108.00)
Premium Gas		43.28	(39.18)	38.38	(45.26)	47.45	(51.38)
Price (pesos/liter)							
Regular Gas		16.68	(1.06)	16.80	(0.70)	16.78	(1.04)
Premium Gas		18.27	(1.00)	18.36	(0.77)	18.46	(1.09)

Panel B: Distribution Stations							
BRAND		South		North		Control (550-750 km)	
		N	%	N	%	N	%
Esso		6	6.9	14	9.1	62	11.0
Oil Com.		1	1.1	7	4.5	47	8.3
Other		1	1.1	1	0.6	30	5.3
Petrobr		11	12.6	15	9.7	22	3.9
Shell		6	6.9	14	9.1	95	16.8
White		23	26.4	46	29.9	135	23.9
YPF		39	44.8	57	37.0	173	30.7
Total		87	100.0	154	100.0	564	100.0

Panel C: Number stations around							
DISTANCE		South		North		Control (550-750 km)	
		$r = 1$	$r = 5$	$r = 1$	$r = 5$	$r = 1$	$r = 5$
Euclidean	Mean	4.08	11.05	4.15	11.32	3.99	18.01
	(SD)	(3.04)	(8.59)	(3.39)	(13.22)	(3.05)	(19.93)
Driving	Mean	2.16	9.52	1.97	8.58	1.88	12.29
	(SD)	(1.37)	(7.70)	(1.15)	(9.14)	(1.07)	(12.90)
Driving	Mean	1.59	7.10	1.39	6.08	1.29	7.21
	(SD)	(0.79)	(5.93)	(0.66)	(4.85)	(0.59)	(6.24)

Note: This table reports descriptive statistics for gasoline stations in the period 2014 –2017 before the reform for the treated and control groups. The distance on each side of the border is 175 km (≈ 108.7 miles) . Panel A shows average monthly quantities (thousand liters) and prices at the station level. Prices are expressed in December 2015 pesos and are residualized from common seasonal patterns. Panel B reports the distribution of stations by brand. Panel C reports the average number of competitors within different radii r . Distances are measured as straight-line distance using the *geodist* package Picard (2010) and as driving distance/time using the *osrmtime* package Huber & Rust (2016) with OpenStreetMap OSRM. Own calculations based on Ministry of Energy data.

Table A.3: Estimation of Equation 1 for price

	(1)	(2)
Log of price	Regular Gas	Premium Gas
Post $\times$ North (non-exempted)	0.0166 *** (0.0037)	0.0137 *** (0.0043)
Post $\times$ South (exempted)	-0.2455 *** (0.0072)	-0.2463 *** (0.0059)
Constant	2.7746 *** (0.0006)	2.8757 *** (0.0006)
Observations	33699	30321
Number of id	705	624
North	134	116
South	76	67
Control	495	441
R-squared	0.9356	0.9571
Equivalent full pass-through	-0.307	-0.307
Pass-through = $\frac{e^{\hat{\beta}} - 1}{Full}$	70.9%	71.1%
CI (95%)	[74.5- 67.3]	[74.1- 68.1]
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 1, where the dependent variable is the log of the retail price in pesos per liter. Columns correspond to regular and premium gasoline, as indicated in the headings. Regressions are estimated using station-by-month data for the period 2014–2017. $South_i$ ($North_i$) is an indicator for stations located within 175 km (≈ 108.7 miles) south (north) of the tax border; the control group consists of stations located 550–750 km (342–466 miles) from the border. All specifications include station fixed effects and month fixed effects. Standard errors are clustered at the county level. For the treatment indicators, $e^{\hat{\beta}} - 1$ gives the percentage change relative to the control group. Significance codes: ***0.01, **0.05, *0.10.

Table A.4: Estimation of Equation 1 for quantity

	(1)	(2)
Log of quantity	Regular Gas	Premium Gas
Post $\times$ North (non-exempted)	-0.0759 *** (0.0232)	-0.1201 *** (0.0389)
Post $\times$ South (exempted)	0.1422 ** (0.0596)	0.3435 *** (0.0461)
Constant	4.1475 *** (0.0042)	3.2403 *** (0.0050)
Observations	33576	30155
Number of id	700	616
North	134	114
South	76	67
Control	490	435
R-squared	0.9384	0.9369
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 1, where the dependent variable is the log of monthly quantity sold (thousand liters). Columns correspond to regular and premium gasoline, as indicated in the headings. The sample period is 2014 – 2017, and the sample is restricted to the balanced panel of stations that operate continuously during this window. $South_i$ ($North_i$) is an indicator for stations located within 175 km (≈ 108.7 miles) south (north) of the tax border; the control group consists of stations located 550 – 750 km (342 – 466 miles) from the border. All regressions include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Coefficients on the treatment indicators can be interpreted as percentage changes in quantities relative to the control group after the reform. Significance codes: ***0.01, **0.05, *0.10.

Table A.5: Legislative Timeline of Law 27,209 (Amendment to Law 23,966)

Date	Event
April 15, 2015	The legislative bill amending Law 23,966 was introduced in Congress.
August 9, 2015	Primary elections were held. The incumbent-party candidate received 38% of the vote, while the main opposition candidate received 24%. No candidate reached the 45% threshold required to win the presidency in the first round.
October 25, 2015	General elections were held. The incumbent-party candidate received 37% of the vote, and the opposition candidate received 34%, triggering a presidential runoff.
November 4, 2015	The bill was approved by the Senate, where the incumbent party held a majority, but had not yet been implemented.
November 22, 2015	The presidential runoff election was held. The opposition candidate won with 51% of the vote, compared to 49% for the incumbent-party candidate.
November 25, 2015	The law was enacted by the federal government, along with other laws and decrees issued by the outgoing administration.
December 1, 2015	Law 27,209 entered into force, and the tax exemption was fully implemented.
December 10, 2015	The new government took office.

Note: Information is based on official records from the Argentine Senate. See the Senate's official website.

Table A.6: Evolution of inflation and gas retail prices (in %)

Year	Annual Inflation	Gasoline increase
2006	10.9	3.1
2007	16.5	-6.6
2008	24.5	20.3
2009	17.4	24.6
2010	23.3	25.0
2011	24.9	21.3
2012	24.9	25.9
2013	24.5	28.6
2014	37.7	52.7
2015	26.0	10.2
2016	39.3	34.2
2017	25.6	21.1
2018	34.3	50.5
2019	53.5	47.5

Note: This table reports annual percentage changes in consumer price inflation and average gasoline retail prices. Inflation is measured using the national consumer price index. Gasoline prices correspond to average retail prices in pesos per liter, expressed in December 2015 pesos. Own calculations based on official inflation statistics and Ministry of Energy price data.

Table A.7: Seasonality in quantities sold for gasoline

Month	Regular gas	Premium gas
February	-0.068 ***	-0.070 ***
March	-0.032 ***	-0.060 ***
April	-0.138 ***	-0.199 ***
May	-0.129 ***	-0.209 ***
June	-0.133 ***	-0.202 ***
July	-0.058 ***	-0.103 ***
August	-0.051 ***	-0.125 ***
September	-0.072 ***	-0.160 ***
October	-0.012 ***	-0.078 ***
November	-0.026 ***	-0.086 ***
December	0.087 ***	0.076 ***
Observations	678061	610069
Number groups	4952	4757
R-squared	0.0028	0.0038
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates from regressions of the log of monthly quantity sold (thousand liters) on month indicators and station fixed effects. The omitted month is January. The sample period is 2008–2022 and includes all gasoline stations in the administrative data. Standard errors are clustered at the station level. Own calculations based on Ministry of Energy data.

Table A.8: Estimates Price for bin regression

	(1)	(2)
Log of price	Regular Gas	Premium Gas
Post $\times$ South bin 175-140km	-0.2794 *** (0.0047)	-0.2664 *** (0.0039)
Post $\times$ South bin 140-105km	-0.2363 *** (0.0043)	-0.2468 *** (0.0036)
Post $\times$ South bin 105-70km	-0.2478 *** (0.0046)	-0.2504 *** (0.0041)
Post $\times$ South bin 70-35km	-0.2374 *** (0.0042)	-0.2291 *** (0.0037)
Post $\times$ South bin 35-0km	-0.2169 *** (0.0064)	-0.2362 *** (0.0057)
Post $\times$ North bin 0-35km	-0.0147 ** (0.0068)	-0.0183 *** (0.0064)
Post $\times$ North bin 35-70km	0.0168 *** (0.0045)	0.0095 ** (0.0042)
Post $\times$ North bin 70-105km	0.0170 *** (0.0033)	0.0200 *** (0.0029)
Post $\times$ North bin 105-140km	0.0178 *** (0.0027)	0.0116 *** (0.0023)
Post $\times$ North bin 140-175km	0.0213 *** (0.0034)	0.0202 *** (0.0031)
Observations	33699	30325
Number of id	713	666
R-squared	0.9314	0.9540
P-value south all equal	0.0000	0.0000
P-value north all equal	0.0001	0.0000

Clustered standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Note: This table reports coefficient estimates and standard errors from regressions of Equation 1, where the dependent variable is the log of the retail price in pesos per liter. Columns correspond to regular and premium gasoline, as indicated in the headings. Regressions are estimated using station-by-month data for the period 2014–2017. $South_i$ ($North_i$) is an indicator for stations located within 175 km (≈ 108.7 miles) south (north) of the tax border; the control group consists of stations located 550–750 km (342–466 miles) from the border. All specifications include station fixed effects and month fixed effects. Standard errors are clustered at the county level. For the treatment indicators, $e^{\hat{\beta}} - 1$ gives the percentage change relative to the control group. Significance codes: ***0.01, **0.05, *0.10.

Table A.9: Estimates Quantity for bin regression

	(1)	(2)
Log of quantity	Regular Gas	Premium Gas
Post $\times$ South bin 175-140km	0.2379 *** (0.0225)	0.3440 *** (0.0252)
Post $\times$ South bin 140-105km	-0.0842 *** (0.0208)	0.1776 *** (0.0231)
Post $\times$ South bin 105-70km	0.1685 *** (0.0221)	0.4057 *** (0.0262)
Post $\times$ South bin 70-35km	0.1873 *** (0.0201)	0.3889 *** (0.0234)
Post $\times$ South bin 35-0km	0.3052 *** (0.0306)	0.5291 *** (0.0364)
Post $\times$ North bin 0-35km	-0.0724 ** (0.0327)	-0.2555 *** (0.0412)
Post $\times$ North bin 35-70km	-0.1559 *** (0.0218)	-0.2872 *** (0.0269)
Post $\times$ North bin 70-105km	-0.0513 *** (0.0160)	-0.1158 *** (0.0187)
Post $\times$ North bin 105-140km	-0.0872 *** (0.0129)	-0.0886 *** (0.0148)
Post $\times$ North bin 140-175km	-0.0409 ** (0.0162)	-0.0656 *** (0.0198)
Observations	33576	30159
Number of id	713	665
R-squared	0.0694	0.2275
P-value south all equal	0.0000	0.0000
P-value north all equal	0.0002	0.0000
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 2 with the log of monthly quantity sold (thousand liters) as the dependent variable. Stations are grouped into 5 distance bins of width 35 km (≈ 22 miles) based on driving time to the closest competitor across the tax border. Each coefficient measures the post-reform change in quantities for stations in a given bin relative to the control group of stations located 550 – 750 km (342 – 466 miles) from the border. All specifications include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Significance codes: ***0.01, **0.05, *0.10.

Table A.10: Test homogeneous pass-through p-values

Number of bins	Regular gas		Premium gas		Extension
	South	North	South	North	
2	0.1356	0.1558	0.0143	0.6919	87.5km
3	0.0603	0.3399	0.0004	0.3041	58.3km
4	0.1253	0.0056	0.0005	0.0883	43.8km
5	0.0111	0.0000	0.0000	0.0954	35.0km
6	0.0000	0.0000	0.0000	0.0000	29.2km
7	0.0001	0.0824	0.0000	0.4002	25.0km
8	0.0003	0.0418	0.0000	0.0432	21.9km
9	0.0000	0.0046	0.0000	0.0000	19.4km
10	0.0000	0.0000	0.0000	0.1131	17.5km

Note: This table reports p-values for the test of homogeneous pass-through for Equation 2. H_0 assumes $\forall b, b' \in Region_r : \beta_b = \beta_{b'}$. Rejecting the null hypothesis means at least one coefficient is different from the others in the region. The areas for each group are defined as follows: 140 km ($\approx$ 87.0 miles) north and south of the border. The control group consists of gas stations 550-750 km (342- 466 miles) located north of the border.

Table A.11: Regression per brand

	(1)	(2)
Log of quantity	Regular Gas	Premium Gas
Post × North (non-exempted)		
× YPF	-0.1412 *** (0.0280)	-0.1107 ** (0.0479)
× Shell	-0.0271 (0.0661)	0.0068 (0.0579)
× White	0.0436 (0.0455)	-0.0699 (0.0756)
× Esso	-0.1652 (0.1075)	-0.2302 * (0.1324)
× Petrobras	-0.0953 ** (0.0396)	-0.2178 *** (0.0602)
× OilComb	-0.0206 (0.0637)	-0.3001 *** (0.0803)
× Other	-0.1737 *** (0.0130)	-0.0345 * (0.0194)
Post × South (exempted)		
× YPF	0.0568 (0.0622)	0.3694 *** (0.0481)
× Shell	0.2115 *** (0.0720)	0.2661 *** (0.0280)
× White	0.2682 ** (0.1163)	0.4682 *** (0.1455)
× Esso	0.2516 * (0.1453)	0.2876 ** (0.1383)
× Petrobras	0.1118 * (0.0634)	0.1835 * (0.0981)
× OilComb	0.0891 *** (0.0129)	0.2967 *** (0.0194)
Constant	4.1475 *** (0.0040)	3.2401 *** (0.0051)
Observations	33576	30155
Number of id	700	616
R-squared	0.9388	0.9372
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 4 for the log of monthly quantity sold (thousand liters), estimated with brand interacted with the treatment indicators. Each column corresponds to a different gasoline, as indicated in the headings. The sample is the balanced panel of stations around the tax border for the period 2014 –2017 . All regressions include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Significance codes: ***0.01, **0.05, *0.10.

Table A.12: Regression per brand

	(1)	(2)
Log of price	Regular Gas	Premium Gas
Post $\times$ North (non-exempted)		
$\times$ YPF	0.0357 *** (0.0037)	0.0321 *** (0.0041)
$\times$ Shell	0.0102 * (0.0058)	0.0042 (0.0050)
$\times$ White	-0.0017 (0.0049)	-0.0057 (0.0058)
$\times$ Esso	0.0134 * (0.0072)	-0.0071 (0.0149)
$\times$ Petrobras	0.0085 (0.0052)	0.0020 (0.0051)
$\times$ OilComb	-0.0165 (0.0108)	-0.0187 (0.0139)
$\times$ Other	0.0003 (0.0021)	0.0048 ** (0.0022)
Post $\times$ South (exempted)		
$\times$ YPF	-0.2310 *** (0.0034)	-0.2303 *** (0.0041)
$\times$ Shell	-0.2446 *** (0.0078)	-0.2542 *** (0.0087)
$\times$ White	-0.2587 *** (0.0202)	-0.2635 *** (0.0205)
$\times$ Esso	-0.2670 *** (0.0090)	-0.2542 *** (0.0121)
$\times$ Petrobras	-0.2580 *** (0.0123)	-0.2632 *** (0.0070)
$\times$ OilComb	-0.2502 *** (0.0021)	-0.2380 *** (0.0022)
Constant	2.7746 *** (0.0005)	2.8757 *** (0.0006)
Observations	33699	30321
Number of id	705	624
R-squared	0.9359	0.9575
Equivalent full pass-through	-0.307	-0.307
Pass-through = $\frac{e^{\beta}-1}{F_{full}}$		
YPF	67.2%	67.0%
CI (%)	[65.5 – 68.9]	[64.9 – 69.1]
Shell	70.7%	73.1%
CI (%)	[66.7 – 74.6]	[68.7 – 77.4]
White	74.3%	75.5%
CI (%)	[64.1 – 84.1]	[65.2 – 85.4]
Esso	76.4%	73.1%
CI (%)	[71.9 – 80.7]	[67.0 – 79.1]
Petrobras	74.1%	75.4%
CI (%)	[67.9 – 80.1]	[71.9 – 78.8]
OilComb	72.1%	69.0%
CI (%)	68 [71.1 – 73.2]	[67.9 – 70.1]
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 4 for the log of the retail price in pesos per liter, estimated with brand interacted with the treatment indicators. Each column corresponds to a different gasoline, as indicated in the headings. The sample is the balanced panel of stations around the tax border for the period 2014 –2017 . All regressions include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Significance codes: ***0.01, **0.05, *0.10.

Table A.13: Distribution of stations before and after reform: Long run

Period	N Firms	South	North	Control (550-750 km)
Pre-period	balanced	89	160	612
	entry	2	9	8
	exit	1	10	28
	total id that appear	92	179	650
Short Run pre and post	balanced	87	154	597
	entry	4	16	23
	exit	1	10	70
	total id that appear	92	179	694

Note: Number of firms located around the tax border: 175 km

Table A.14: Estimates Price for unbalanced sample.

	(1)	(2)
Log of price	Regular Gas	Premium Gas
Post $\times$ North (non-exempted)	-0.0019 (0.0026)	0.0001 (0.0031)
Post $\times$ South (exempted)	-0.2515 *** (0.0042)	-0.2489 *** (0.0049)
Constant	2.7665 (0.0011)	2.8677 (0.0013)
Observations	171652	155386
Number of id	3672	3276
North	2927	2608
South	213	198
Control	532	470
R-squared	0.9447	0.9608
Equivalent full pass-through	-0.307	-0.307
Pass-through = $\frac{e^{\beta}-1}{Full}$	72.5%	71.8%
CI (95%)	[74.5- 70.4]	[74.2- 69.4]
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Table A.15: Estimates Price for bin regression

Note: This table reports coefficient estimates and standard errors from regressions of Equation 1 with the log of the retail price in pesos per liter as the dependent variable, using the unbalanced panel of stations around the tax border. The sample includes all stations that operate at any point during 2014 –2017 , allowing for entry and exit. $South_i$ ($North_i$) is an indicator for stations located within 175 km (≈ 108.7 miles) south (north) of the border; the control group consists of stations located 550 – 750 km (342 – 466 miles) from the border. All regressions include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Significance codes: ***0.01, **0.05, *0.10.

Table A.16: Estimates Quantity for unbalanced sample.

	(1)	(2)
Log of quantity	Regular Gas	Premium Gas
Post $\times$ North (non-exempted)	0.0024 (0.0148)	0.0278 (0.0217)
Post $\times$ South (exempted)	0.0626 ** (0.0286)	0.2882 *** (0.0259)
Constant	4.2992 (0.0065)	3.3059 (0.0095)
Observations	170889	154452
Number of id	3649	3256
North	2909	2593
South	211	197
Control	529	466
R-squared	0.9372	0.9381
Clustered standard errors in parentheses		
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$		

Note: This table reports coefficient estimates and standard errors from regressions of Equation 1 with the log of monthly quantity sold (thousand liters) as the dependent variable, using the unbalanced panel of stations around the tax border. The sample includes all stations that operate at any point during 2014–2017. $South_i$ ($North_i$) is an indicator for stations located within 175 km (≈ 108.7 miles) south (north) of the border; the control group consists of stations located 550 – 750 km (342 – 466 miles) from the border. All specifications include station fixed effects and month fixed effects. Standard errors are clustered at the county level. Significance codes: ***0.01, **0.05, *0.10.

Table A.17: Distribution of stations before and after reform: Long run

Panel A: Number of Stations												
Brand	South				North				Control (550-750 km)			
	Pre				Pre				Pre			
	N	%	N	%	N	%	N	%	N	%	N	%
Axion	9	7.69	12	9.68	13	4.89	20	7.55	53	7.01	66	8.30
Esso	7	5.98	7	5.65	17	6.39	19	7.17	59	7.80	56	7.04
Oil Com.	0	0.00	0	0.00	9	3.38	7	2.64	43	5.69	48	6.04
Other	17	14.53	18	14.52	26	9.77	26	9.81	88	11.64	100	12.58
Petrobr	10	8.55	10	8.06	25	9.40	21	7.92	62	8.20	60	7.55
Shell	11	9.40	13	10.48	27	10.15	27	10.19	119	15.74	126	15.85
White	25	21.37	27	21.77	70	26.32	67	25.28	160	21.16	160	20.13
YPF	38	32.48	37	29.84	79	29.70	78	29.43	172	22.75	179	22.52
Total	117	100.0	124	100.0	266	100.0	265	100.0	756	100.0	795	100.0

Note: This table reports descriptive statistics for gasoline stations in the long run by region (south, north, and control), using the full sample of stations located within 175 km of the tax border. For each region, the table compares the distribution of stations and brands before the reform and 5 years after the reform over the period 2020-2021. Own calculations based on Ministry of Energy administrative data.